

## {3} Current Electricity



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- \* INTRODUCTION
- \* ELECTRIC CURRENT
- \* ELECTRIC CURRENTS IN CONDUCTORS
- \* OHM' S LAW
- \* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY
- \* LIMITATIONS OF OHM 'S LAW



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\* INTRODUCTION

\* ELECTRIC CURRENT

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\* OHM'S LAW

\* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY

\* LIMITATIONS OF OHM'S LAW

\* TEMPERATURE DEPENDENCE OF RESISTIVITY

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\* CELLS IN SERIES AND IN PARALLEL

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- \* **Charges in motion constitute an electric current.**
- \* In the present chapter, we shall study some of the basic laws concerning **steady electric currents**.



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- \* The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.



**Ex 1 : In cosmic rays  $0.15$  protons/cm<sup>2</sup>-s are entering the earth's atmosphere. If the radius of the earth is  $6400$  km, what is the current received by the earth in the form of cosmic rays?**



**Ex 2:** In neon gas discharge tube  $2.9 \times 10^{18}$   $\text{Ne}^+$  ions move to the right through a cross-section of the tube each second, while  $1.2 \times 10^{18}$  electrons move to the left in this time. What is the net electric current in the tube?



**Ex 3 :** In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius  $5 \times 10^{-11}$  m, at a speed of  $2.2 \times 10^6$  m/s. What is the current associated with electron motion?



**Ex 4 :** The current in a wire varies with time according to the relation  $I = 2 \text{ A} + (0.6 \text{ A/s}^2) t^2$ .

- (a) How many coulomb of charge pass a cross-section of the wire in the time interval between  $t = 0$  and  $t = 10 \text{ s}$ ?
- (b) What constant current would transport the same charge in same charge in the same time interval?



# ELECTRIC CURRENTS IN CONDUCTORS

## Electric Currents in Conductors

Electric current is the flow of electric charge through a conductor.

The direction of current flow is defined as the direction in which positive charges would move.

Current is measured in amperes (A).

The relationship between current, voltage, and resistance is given by Ohm's Law:

$$I = \frac{V}{R}$$

where  $I$  is the current,  $V$  is the voltage, and  $R$  is the resistance.

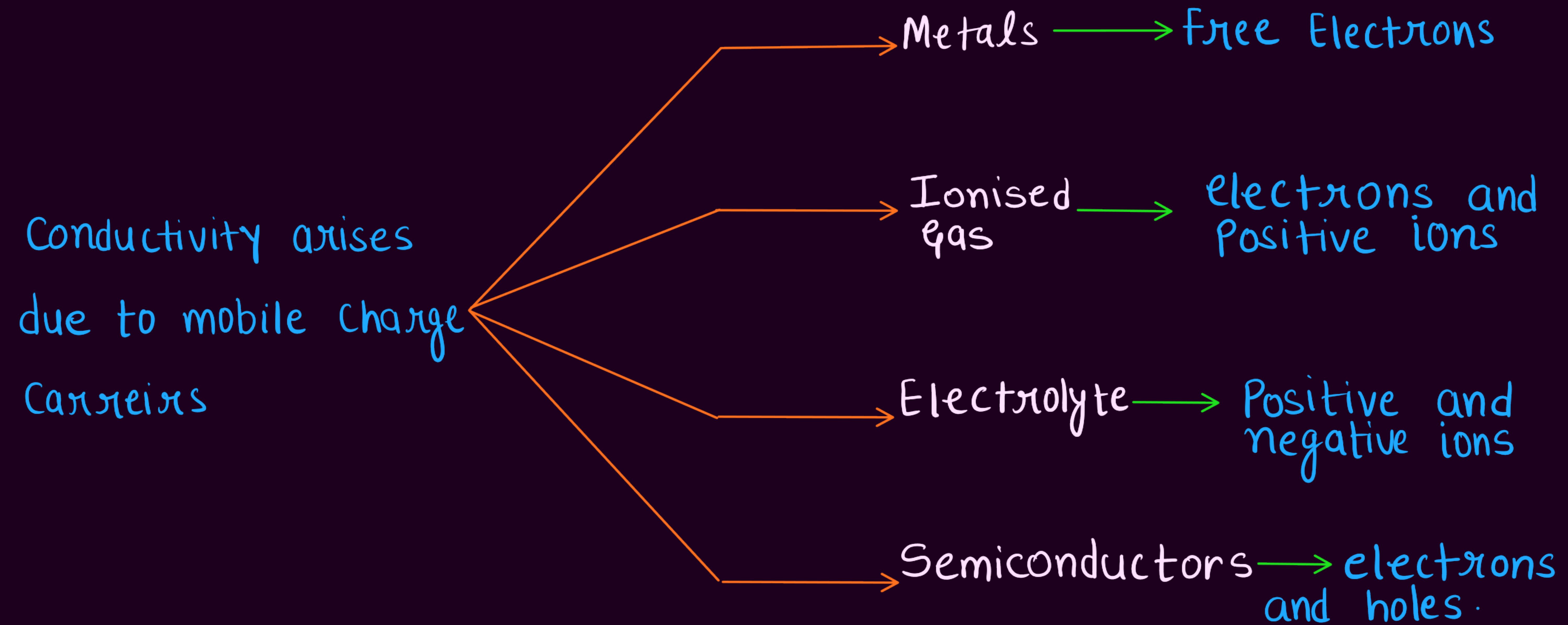
Current flows from the positive terminal of a battery to the negative terminal.

The total current entering a junction is equal to the total current leaving the junction.

Current is a scalar quantity.

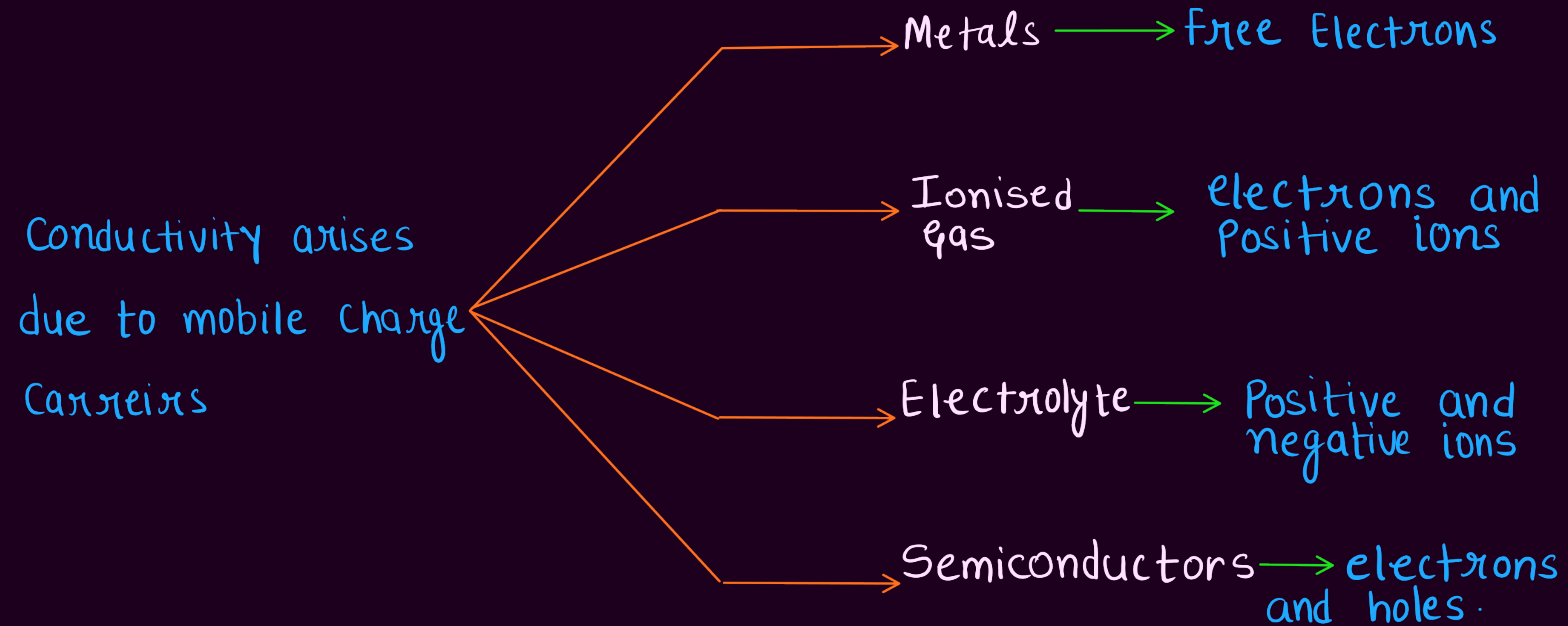


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\* In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.



# ELECTRIC CURRENTS IN CONDUCTORS

## DRIFT VELOCITY

Electrons in a conductor are in constant motion due to thermal energy.

When an electric field is applied, they acquire a net drift velocity.

The drift velocity is the average velocity of electrons in the direction of the electric field.

It is denoted by  $v_d$ .

The drift velocity is much smaller than the thermal velocity.

It is the net velocity of electrons in the direction of the electric field.

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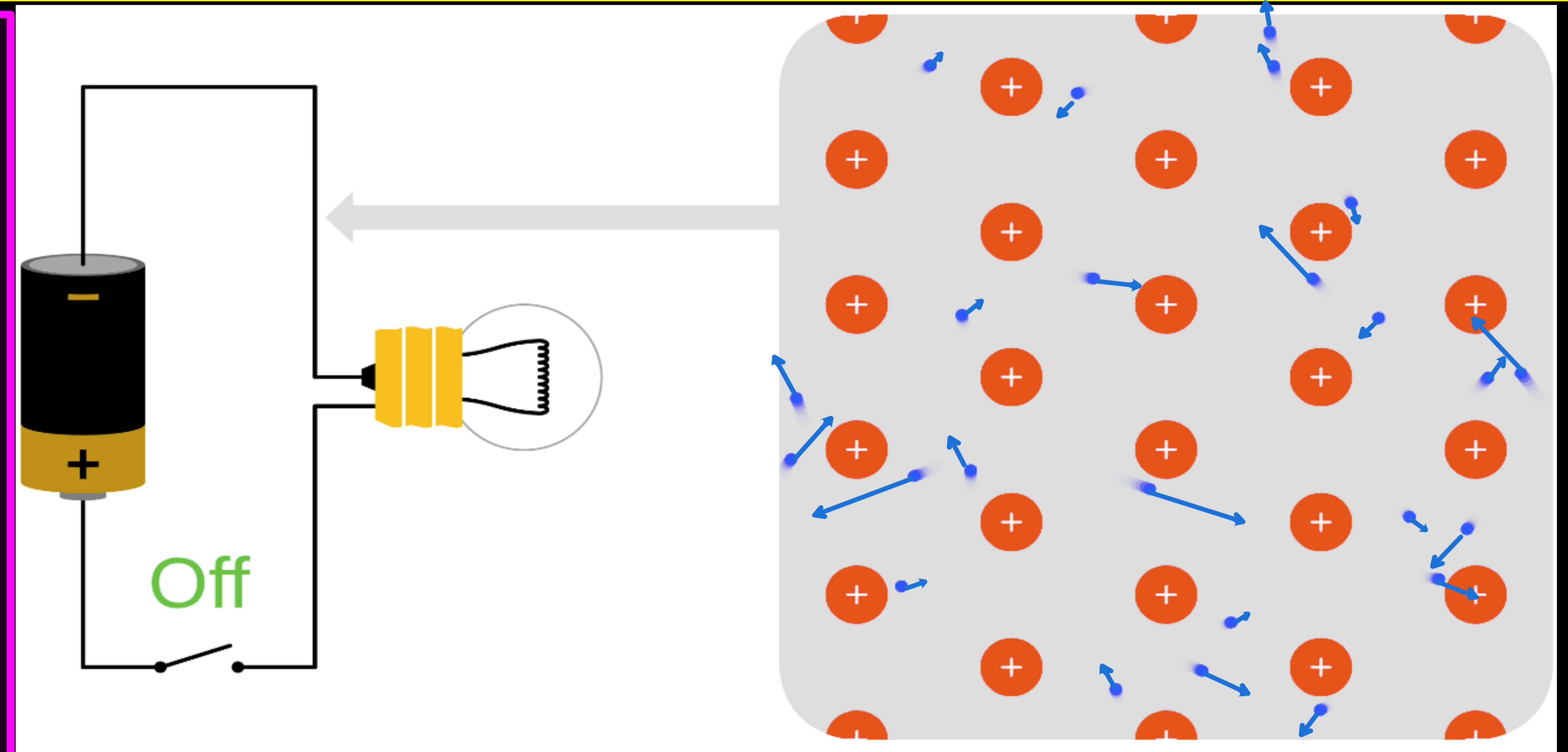
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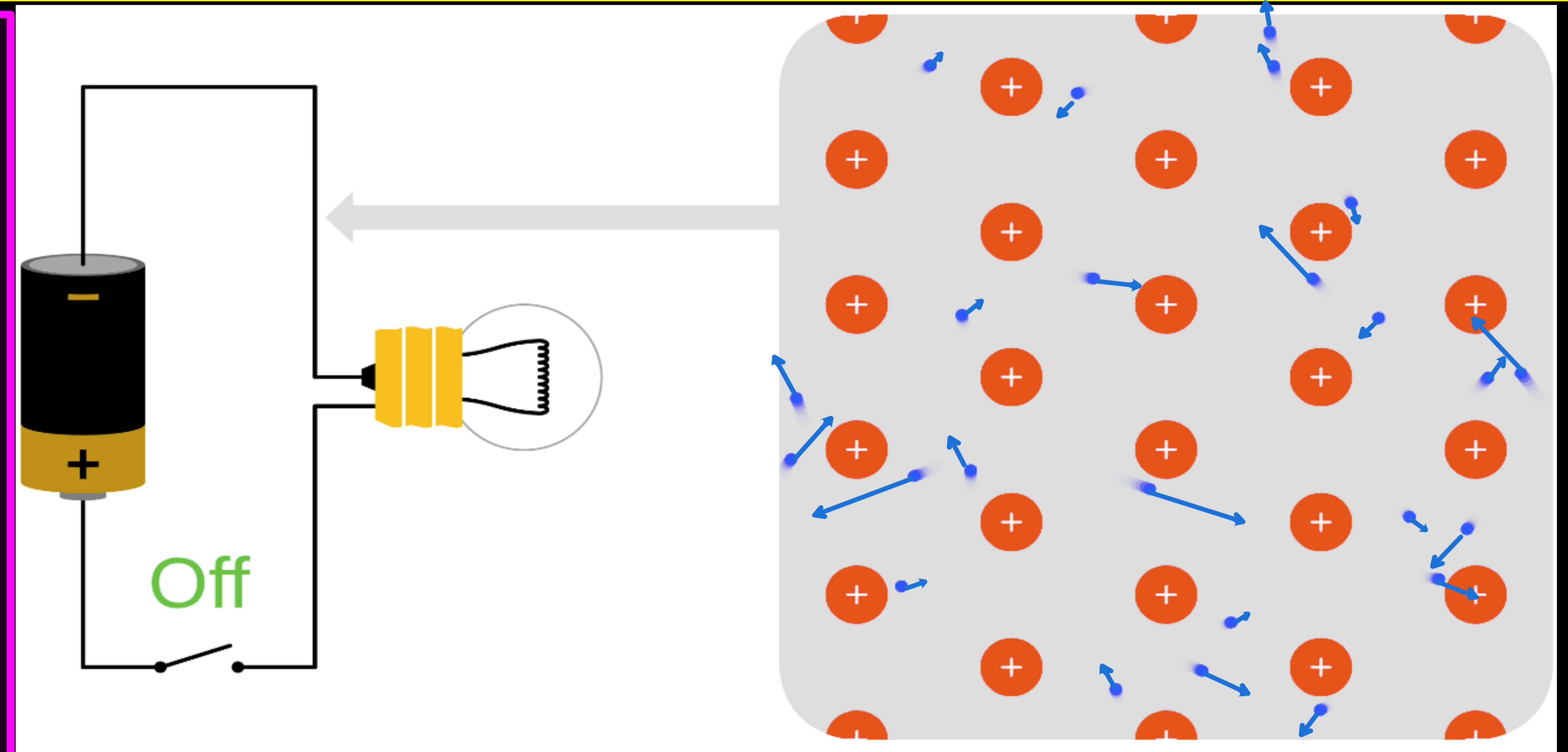
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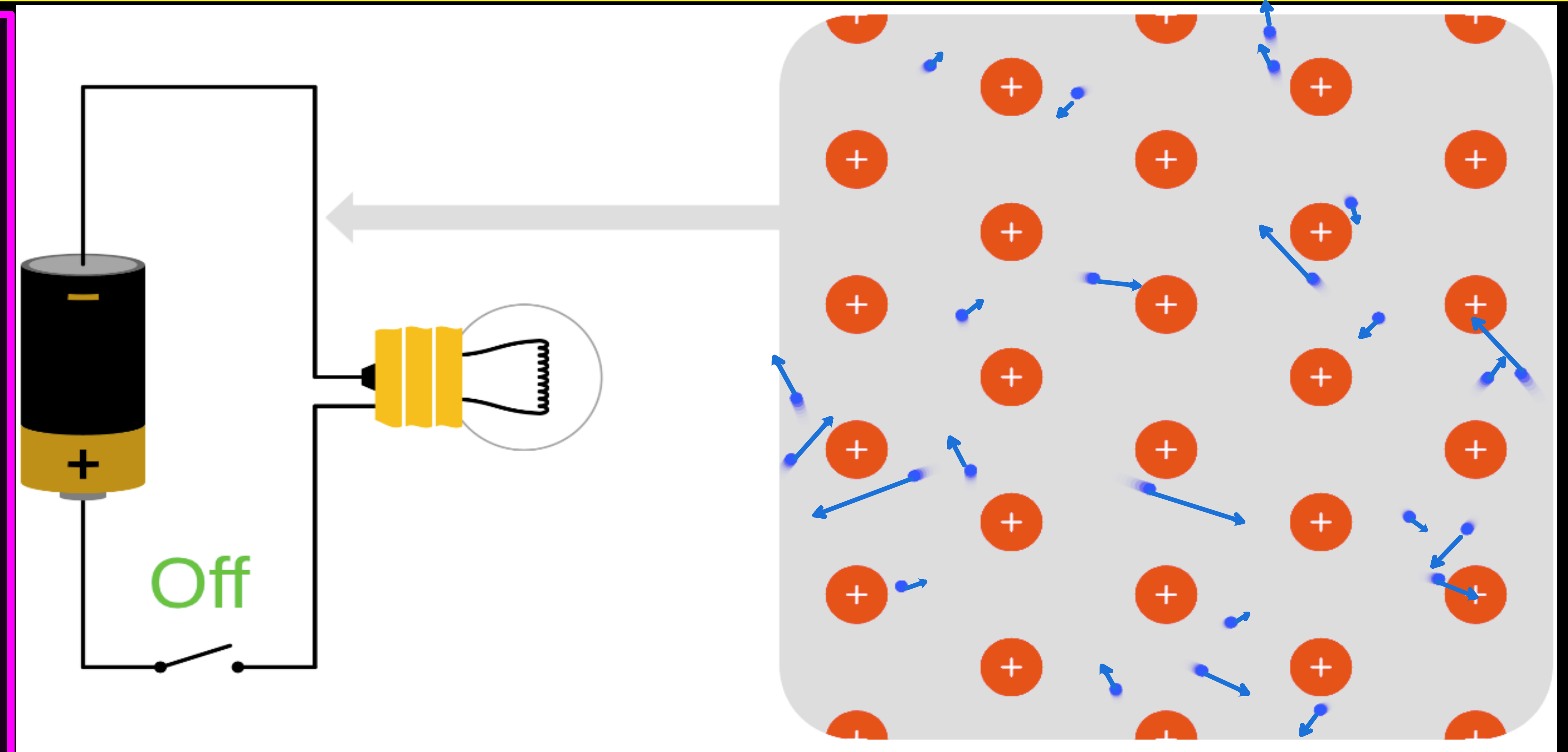




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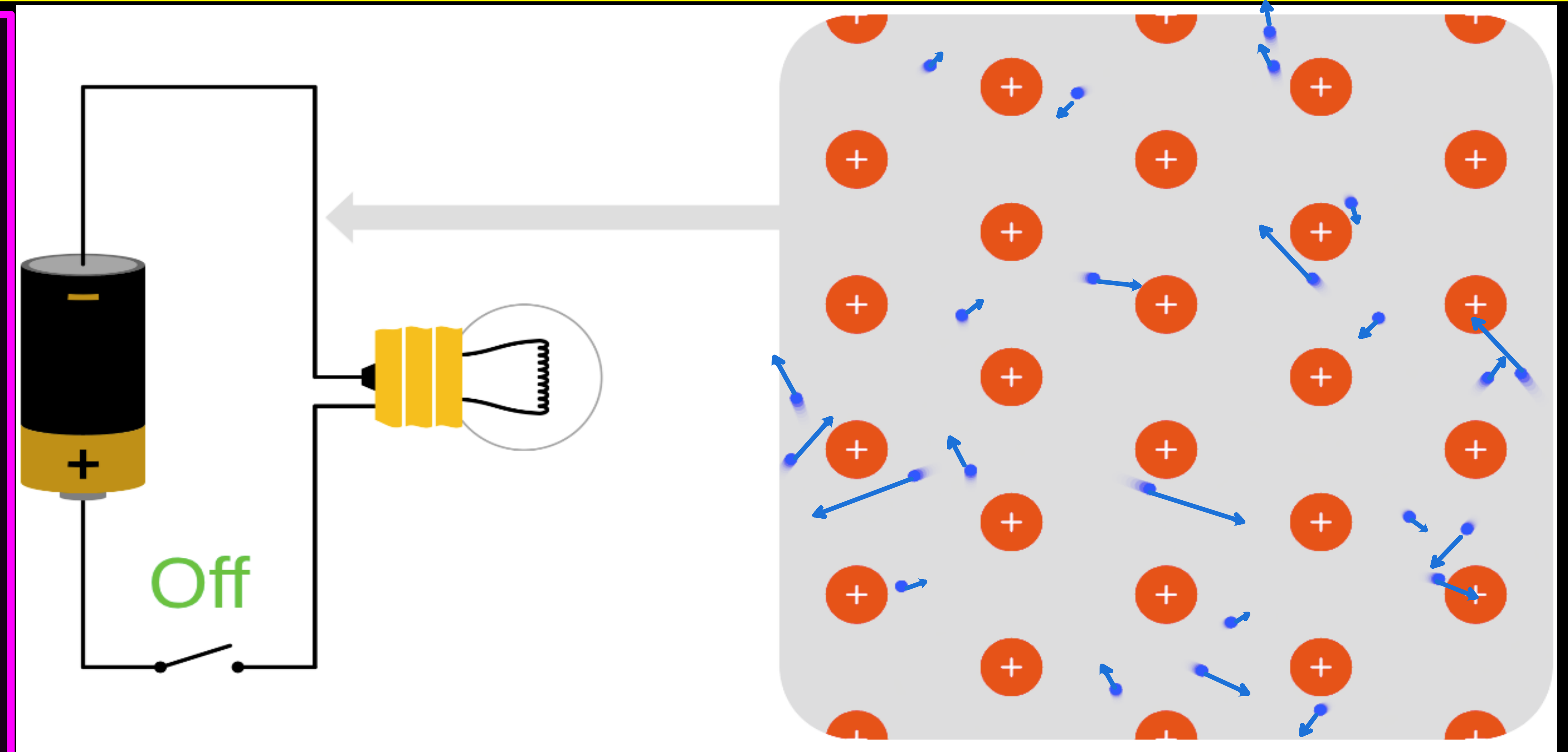


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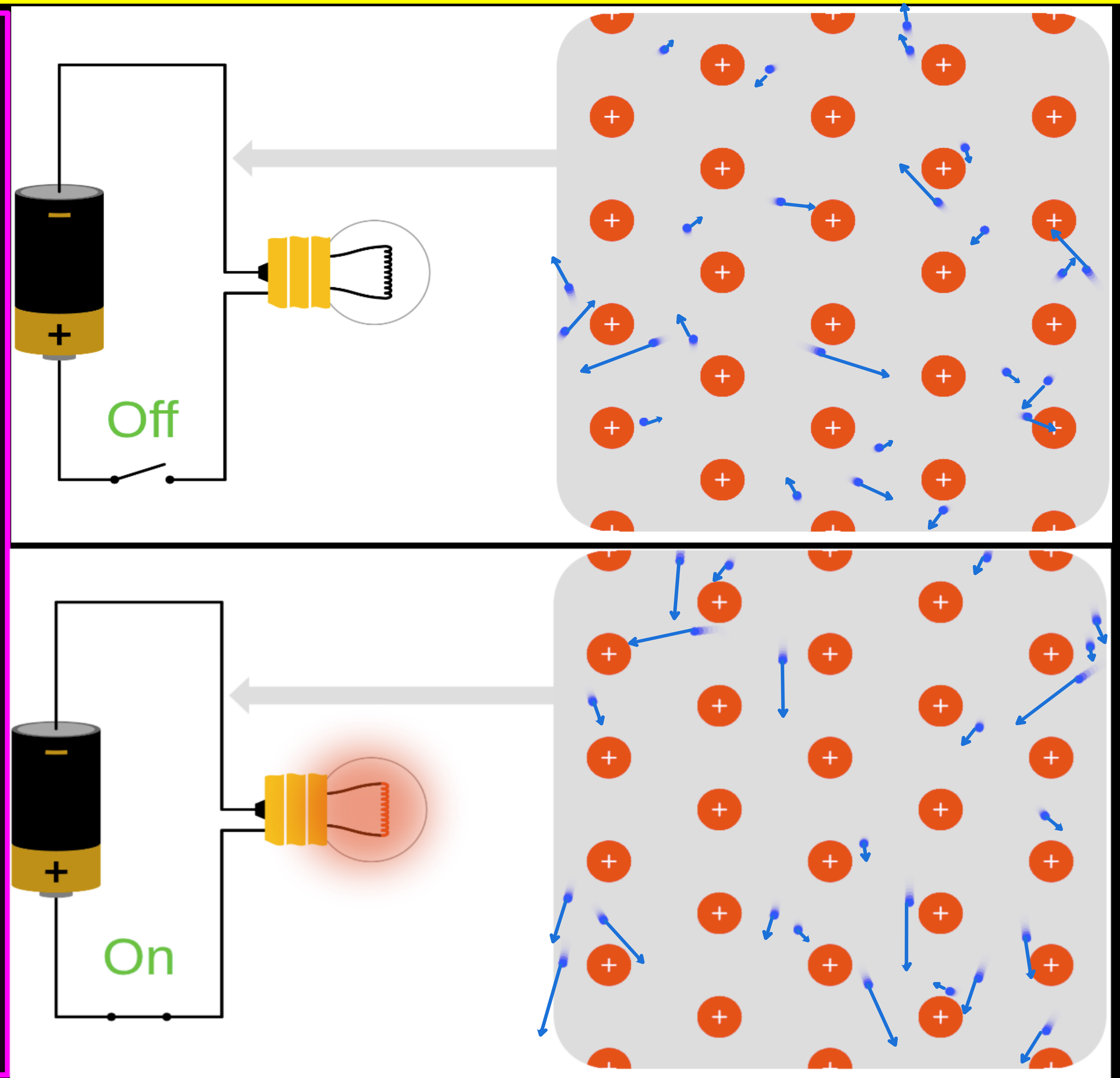
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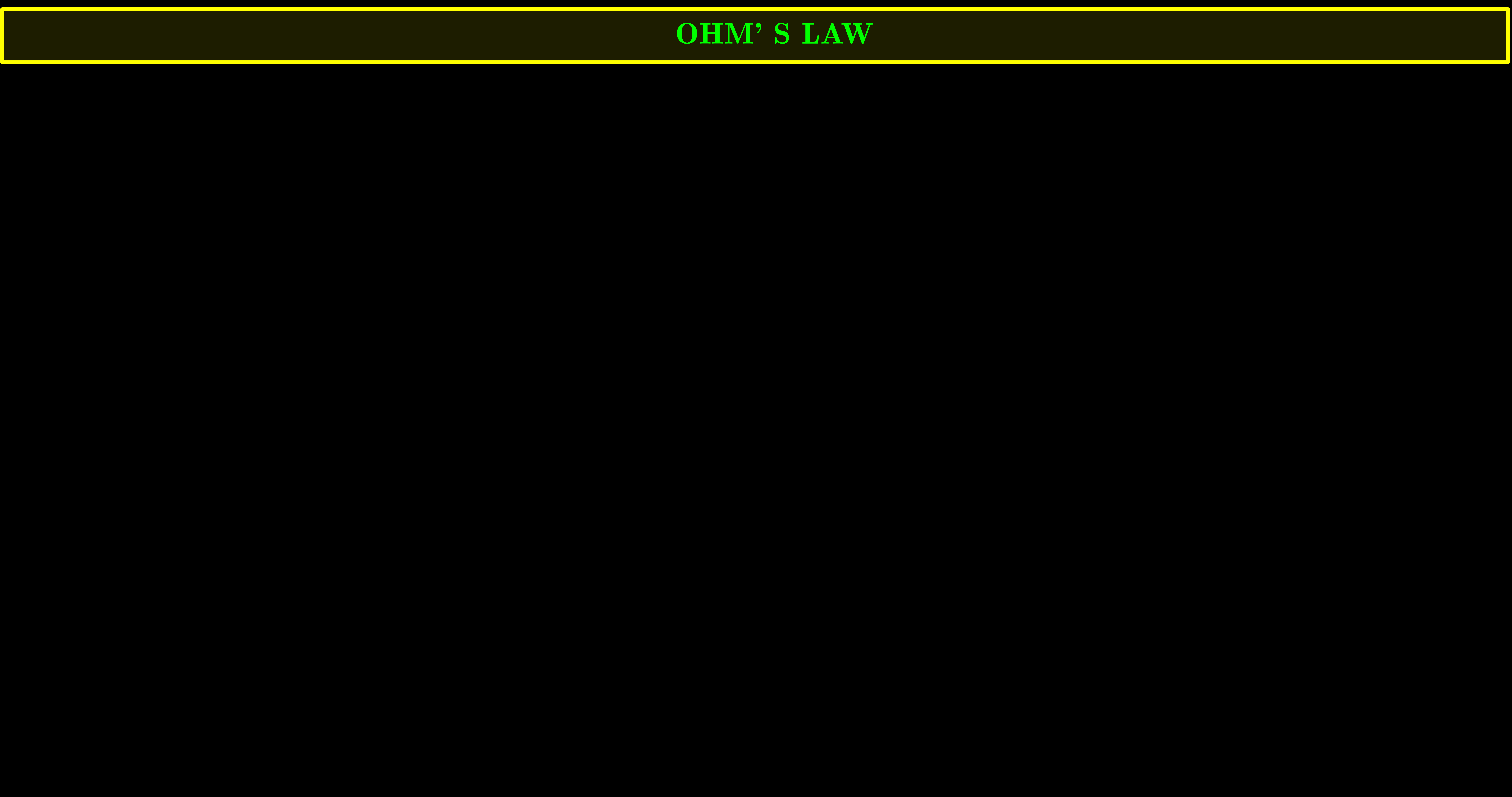
**Case 2 : When electric field is present :**

\* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.





# OHM'S LAW





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\* A basic law regarding flow of currents was discovered by G.S. Ohm in 1828, long before the physical mechanism responsible for flow of currents was discovered.

Imagine a conductor through which a current  $I$  is flowing and let  $V$  be the potential difference between the ends of the conductor. Then Ohm's law states that the current in a conductor is directly proportional to the voltage applied:



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$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

\* where the constant of proportionality called resistivity ( $\rho$ ) depends on the material



# CURRENT DENSITY AND EQUIVALENT FORM OF OHM'S LAW

Consider a rectangular block of material with length  $l$ , cross-sectional area  $A$ , and potential difference  $V$  applied across its ends.

The current  $I$  flowing through the block is given by Ohm's Law:

$$I = \frac{V}{R}$$

where  $R$  is the resistance of the block. The resistance  $R$  is given by:

$$R = \frac{\rho l}{A}$$

where  $\rho$  is the resistivity of the material. Substituting the expression for  $R$  into Ohm's Law, we get:

$$I = \frac{V A}{\rho l}$$

The current density  $J$  is defined as the current per unit area:

$$J = \frac{I}{A}$$

Substituting the expression for  $I$  into the definition of  $J$ , we get:

$$J = \frac{V}{\rho l}$$



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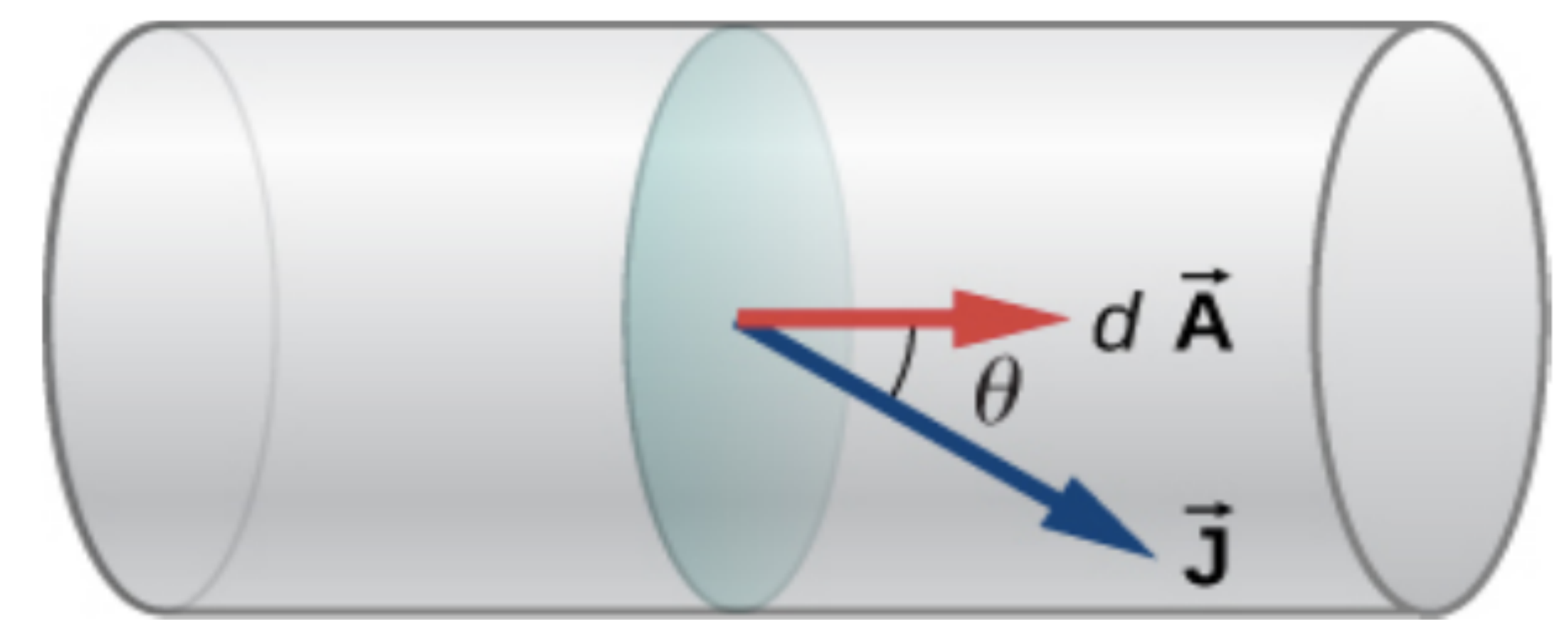


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**Figure 9.12** The current density  $\vec{J}$  is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.



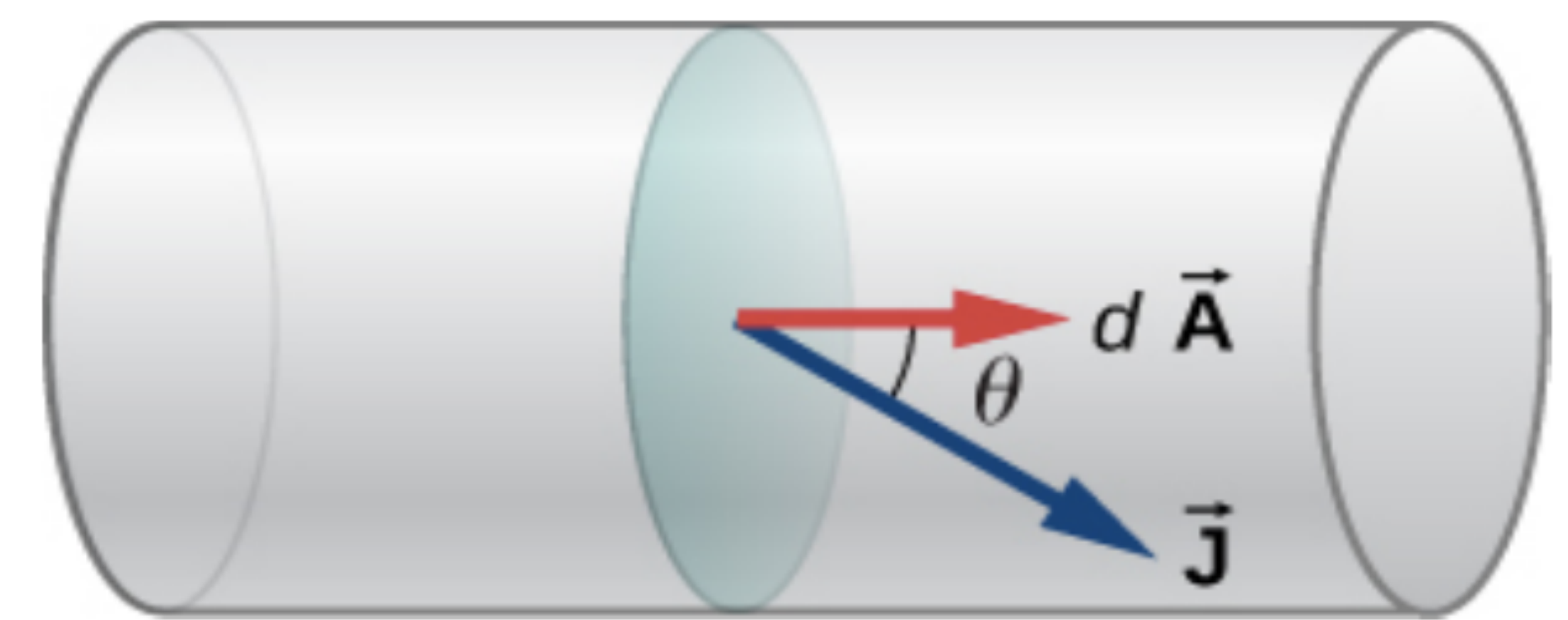
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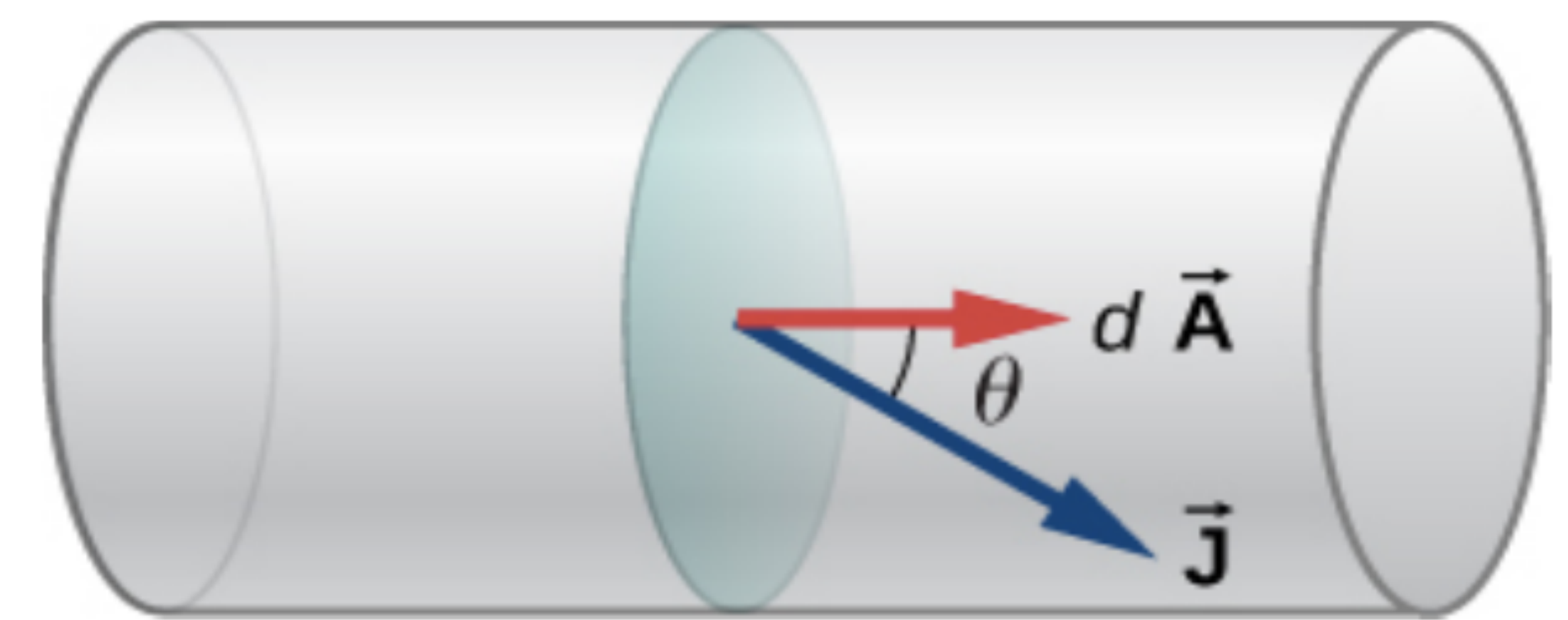
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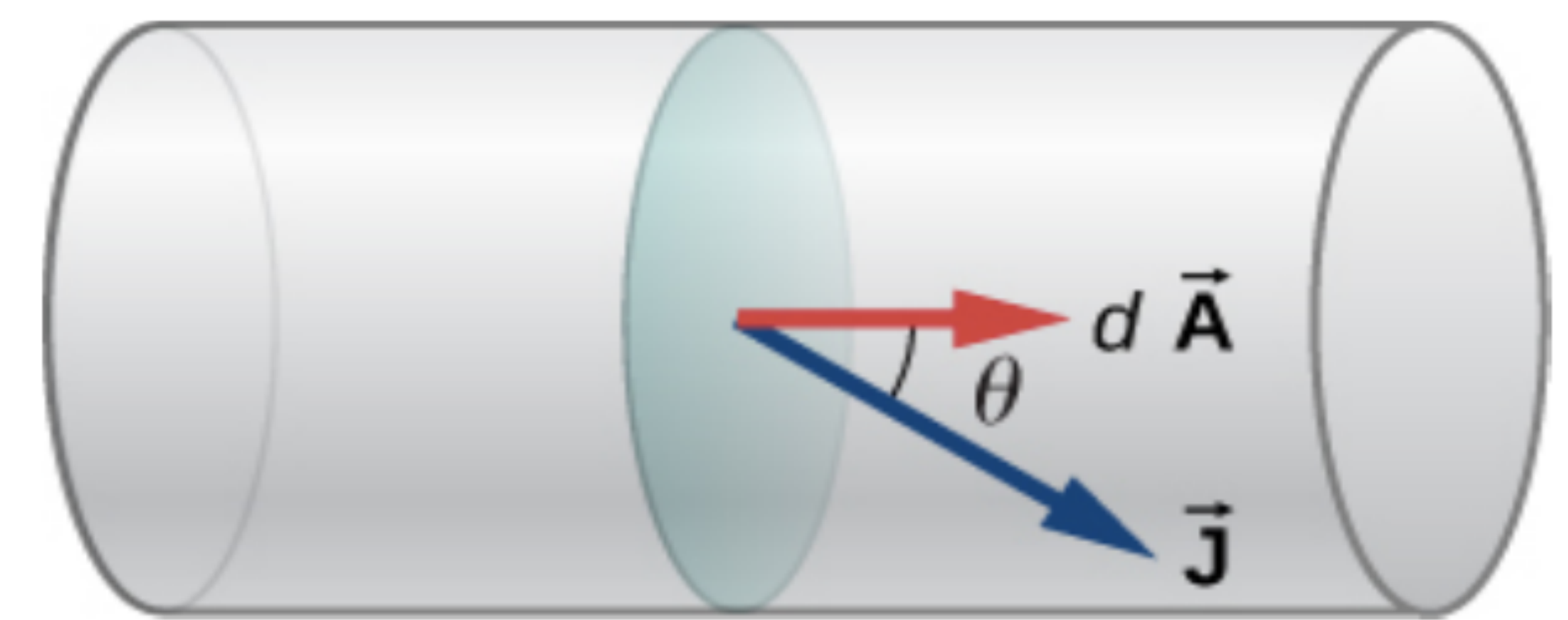
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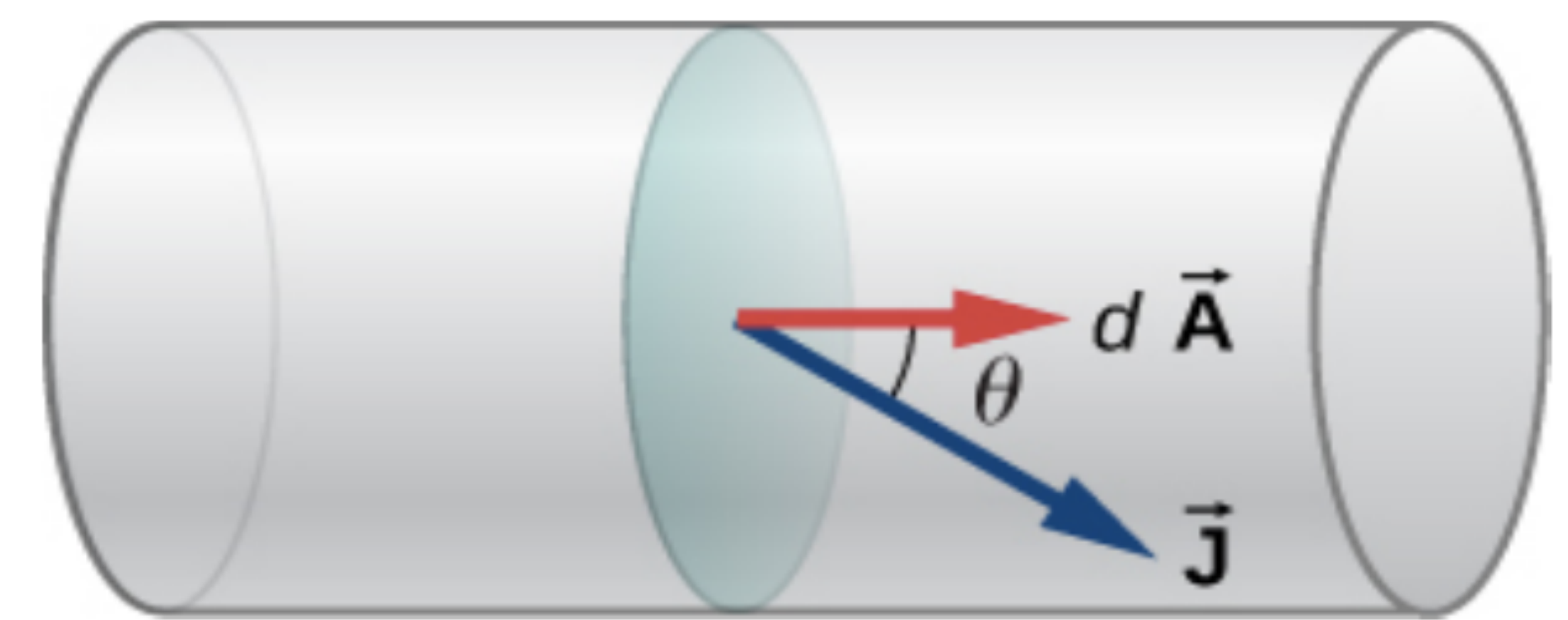
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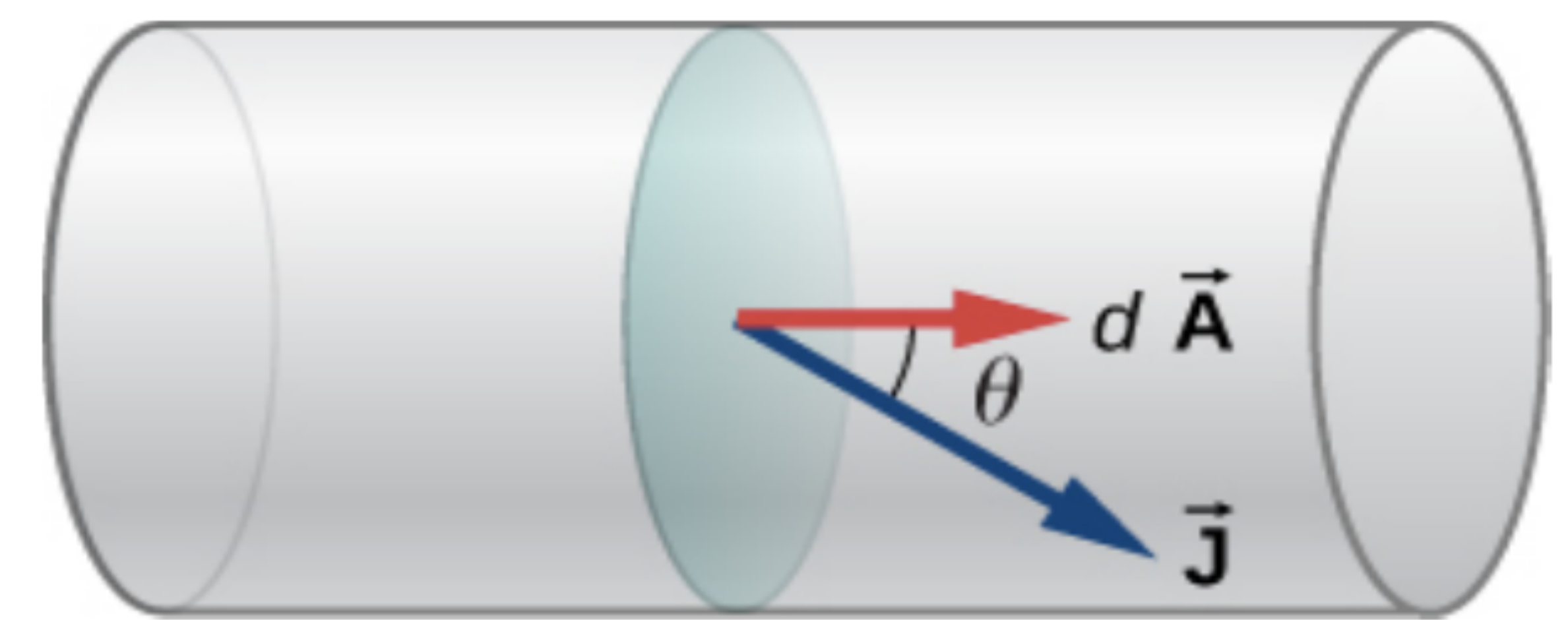
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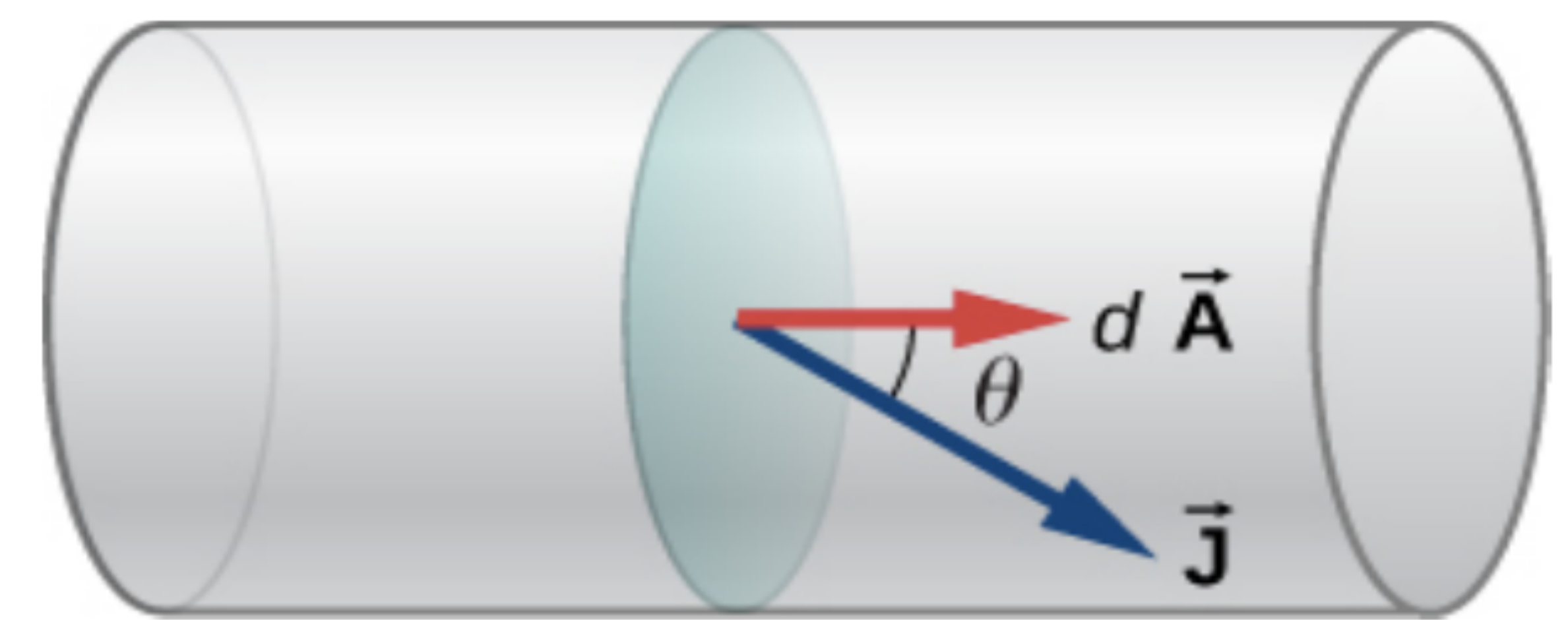
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**3.6** A negligibly small current is passed through a wire of length 15 m and uniform cross-section  $6.0 \times 10^{-7} \text{ m}^2$ , and its resistance is measured to be  $5.0 \text{ } \Omega$ . What is the resistivity of the material at the temperature of the experiment?



**Ex 6 : A wire 50 cm long and 1 mm<sup>2</sup> in cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is**

*a)*  $2 \times 10^{-7} \Omega m$       *b)*  $5 \times 10^{-7} \Omega m$

*c)*  $4 \times 10^{-6} \Omega m$       *d)*  $1 \times 10^{-6} \Omega m$



**Ex 7: A wire of length  $L$  and cross-section area  $A$  has resistance  $R$ . What will be the resistance of the wire if it is stretched to twice its original length? Assume that the density and resistivity of material do not change when the wire is stretched.**



**Ex 8 :** A wire of length  $L$  is drawn such that its diameter is reduced to half of its original diameter. If the initial resistance of the wire is  $10\ \Omega$ , its new resistance would be

a)  $40\ \Omega$

(b)  $80\ \Omega$

(c)  $120\ \Omega$

(d)  $160\ \Omega$

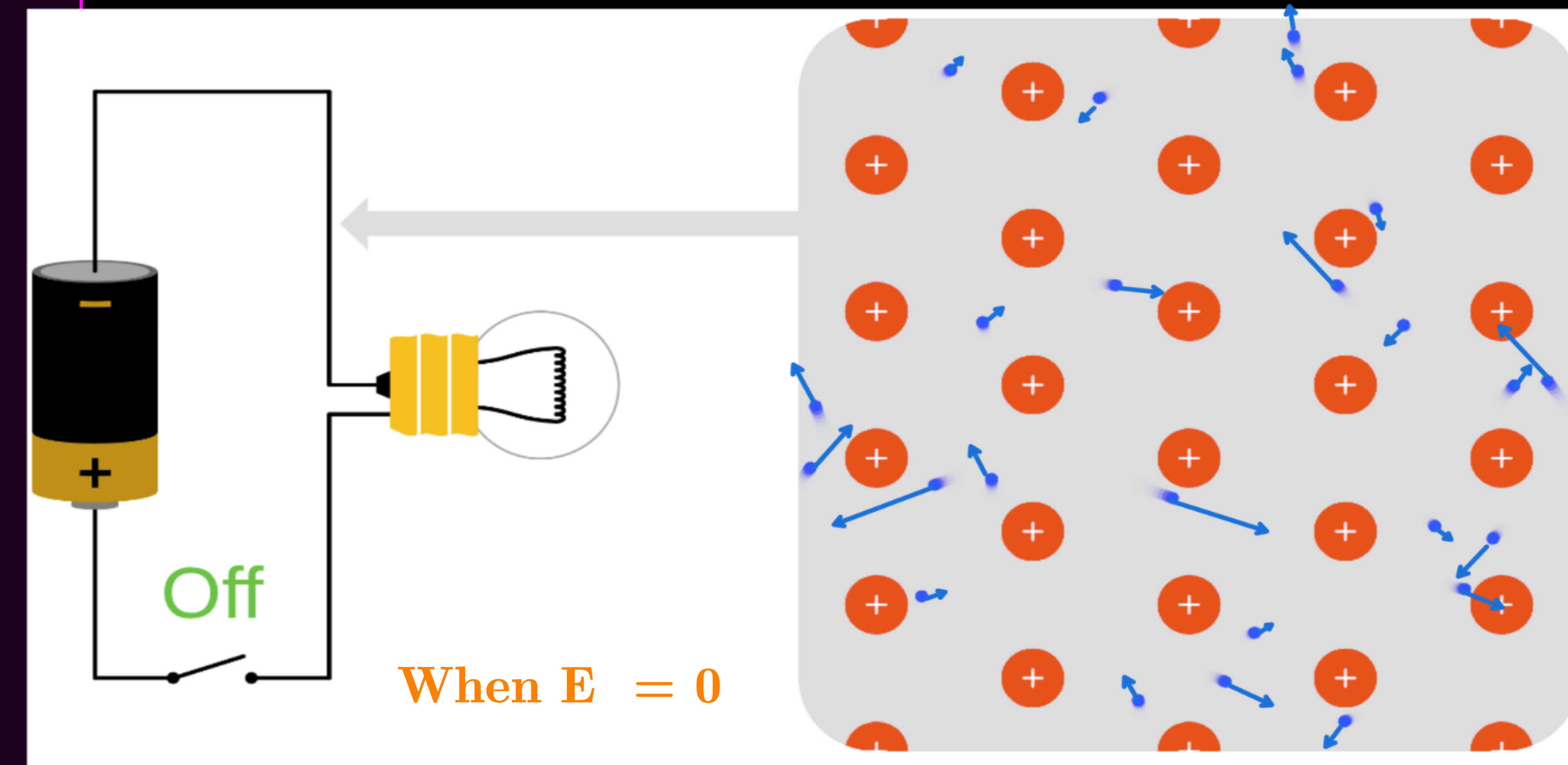


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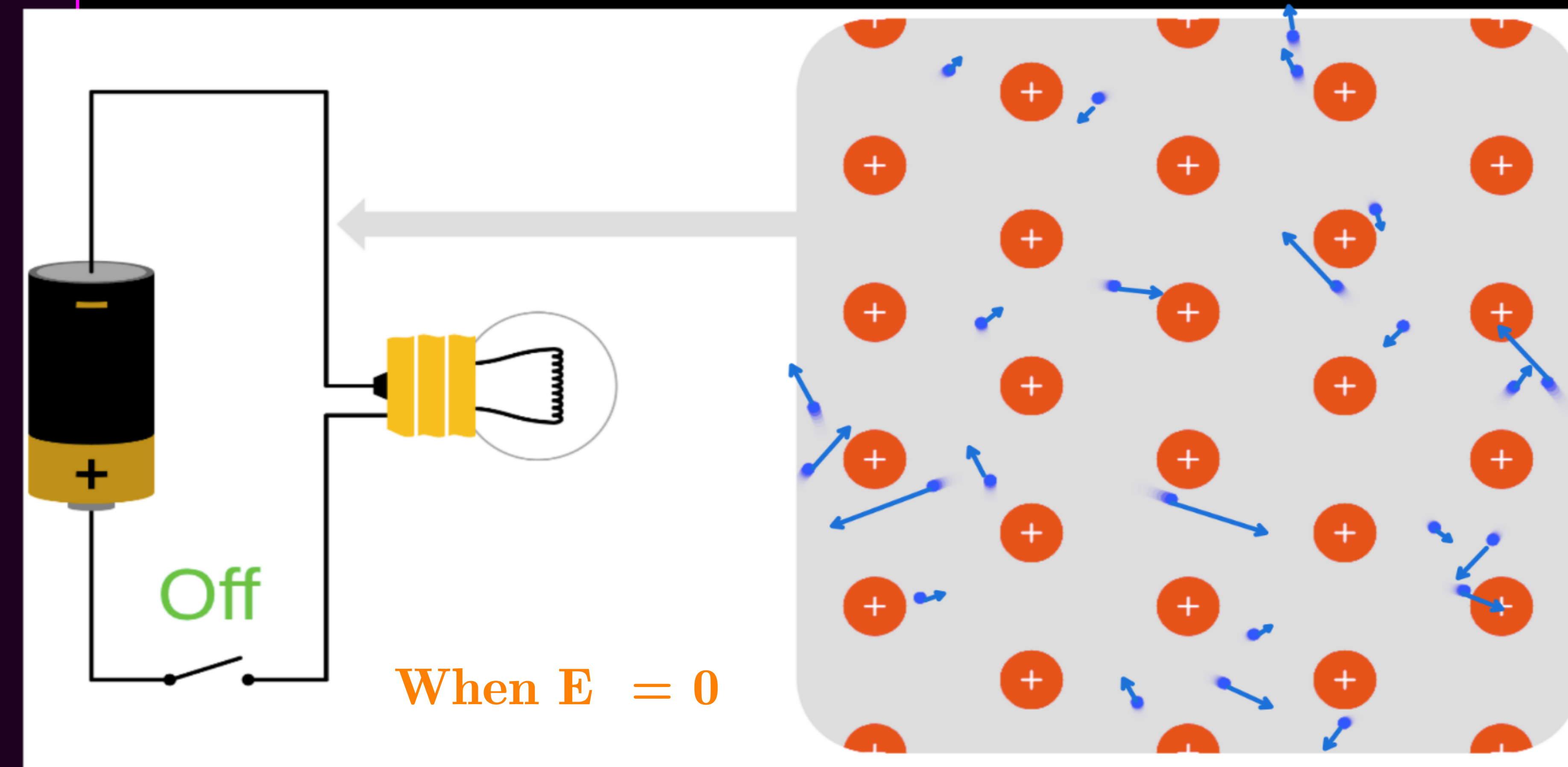




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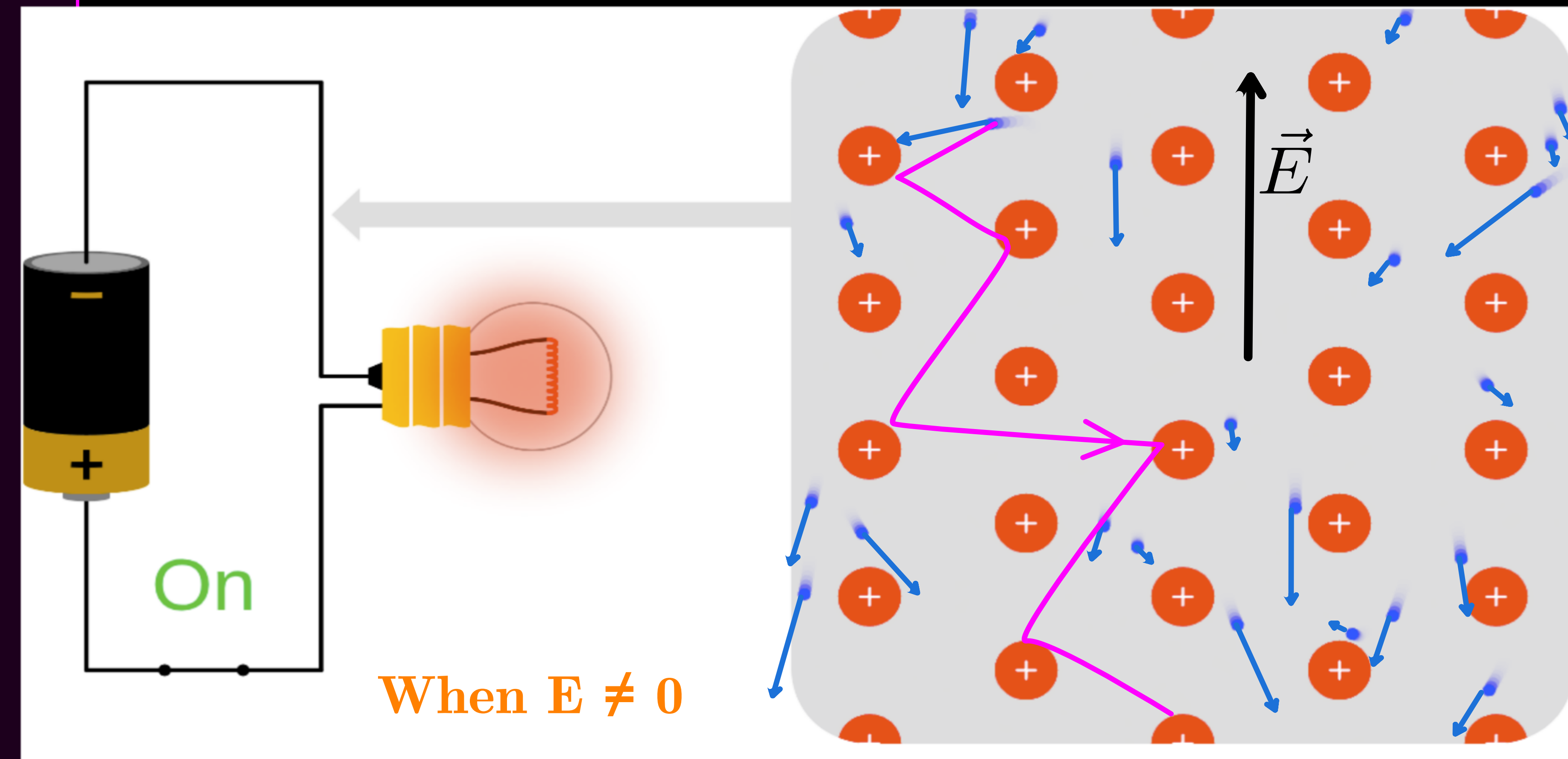
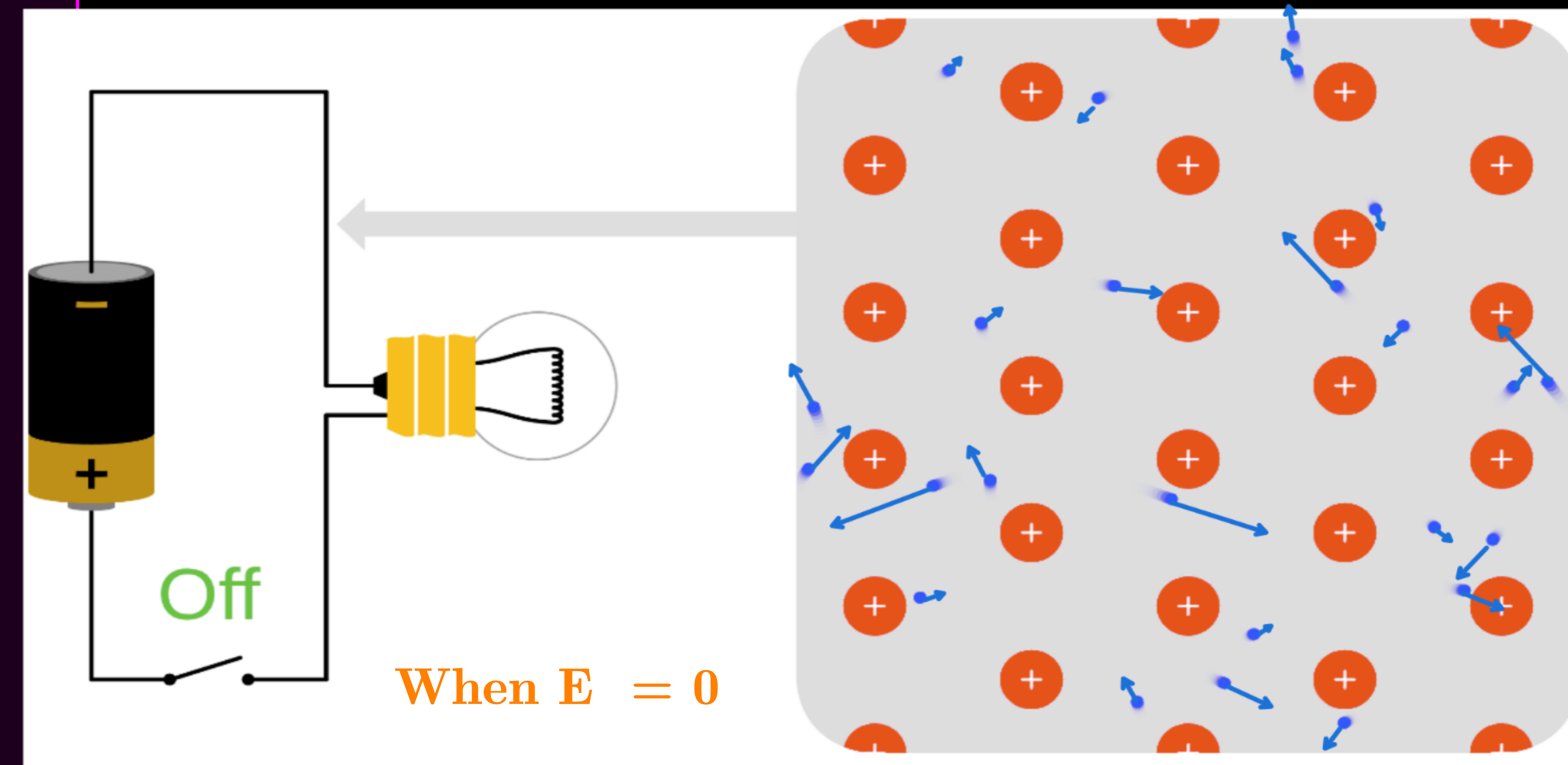
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- \* When an electric field is present. Electrons will experience a force in the opposite direction of the Electric field.

$$\vec{F} = -e\vec{E}$$





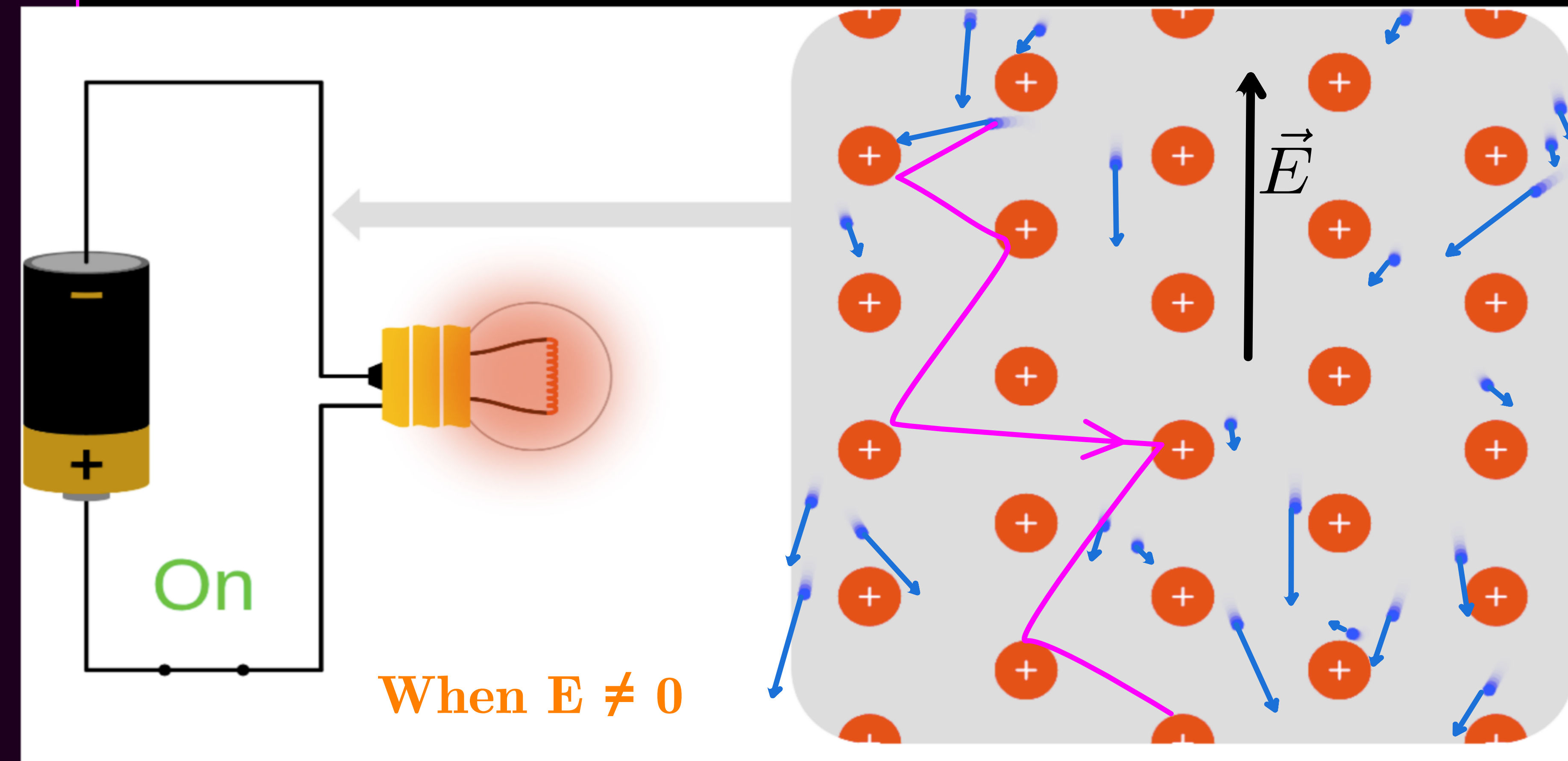
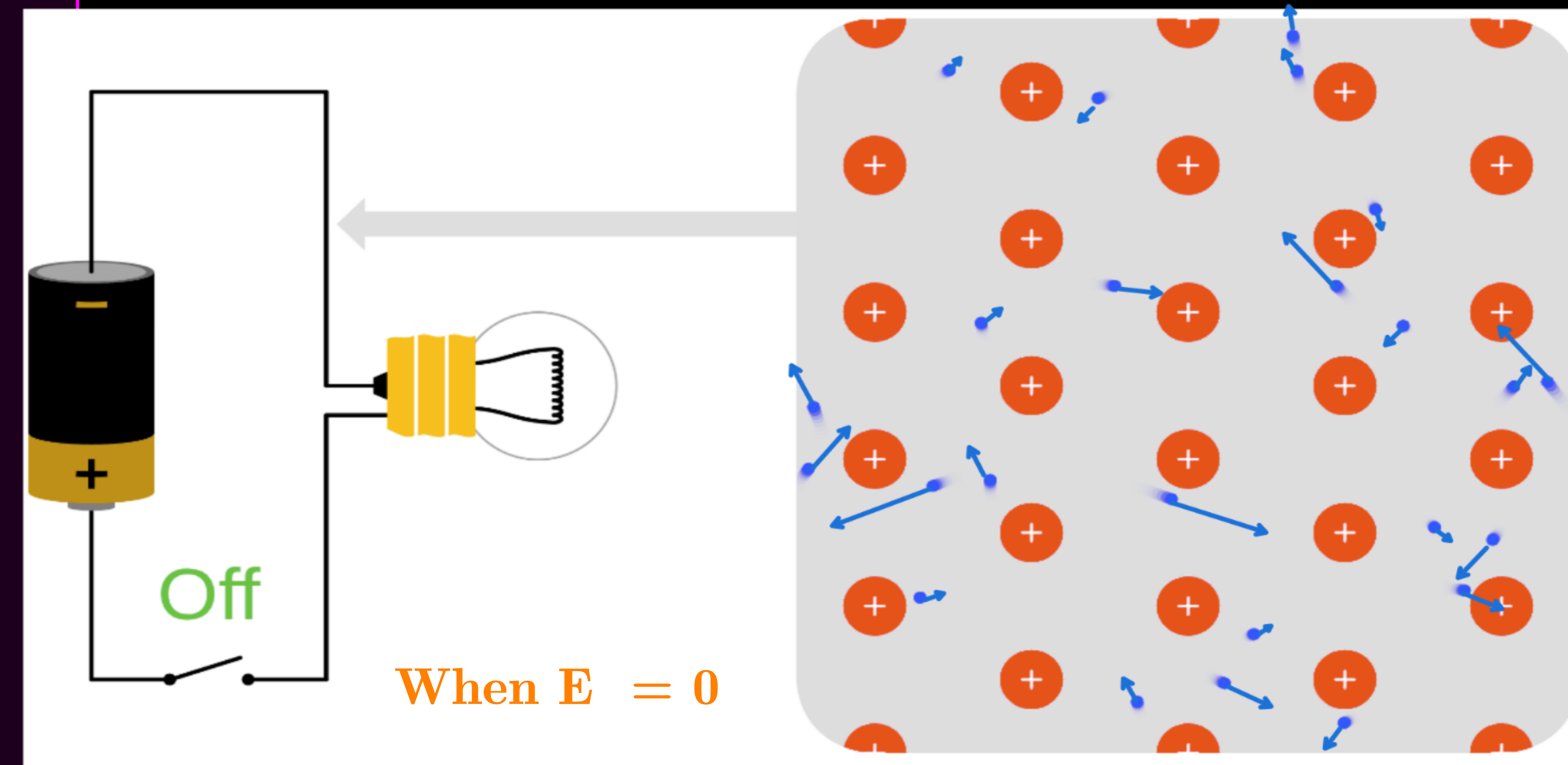
# DRIFT VELOCITY

- \* In metals the charge carriers are negatively charged electrons that move to create an electric current.
- \* Without an applied electric field, electrons move around randomly in a material.
- \* If we consider all the  $n$  electrons with random velocities ( $u_1, u_2, u_3, \dots, u_n$ ), their average velocity will be zero since their directions are random.

$$\vec{u}_{\text{avg.}} = \frac{\vec{u}_1 + \vec{u}_2 + \dots + \vec{u}_n}{n} = \frac{1}{n} \sum_{i=1}^n \vec{u}_i = 0$$

- \* When an electric field is present. Electrons will experience a force in the opposite direction of the Electric field.

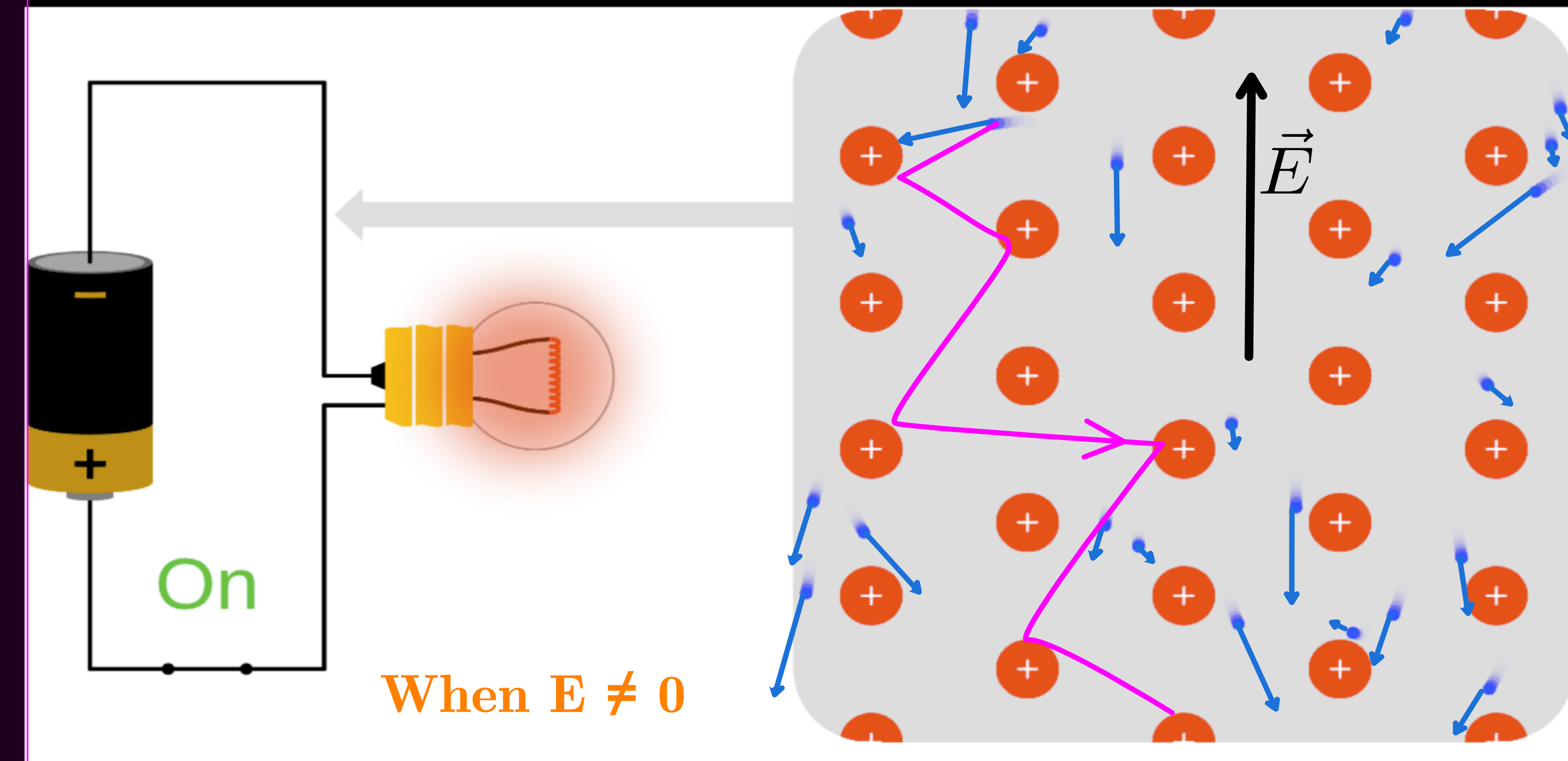
$$\vec{F} = -e\vec{E} \quad \therefore \text{acceleration } \vec{a} = \frac{-e\vec{E}}{m_e}$$





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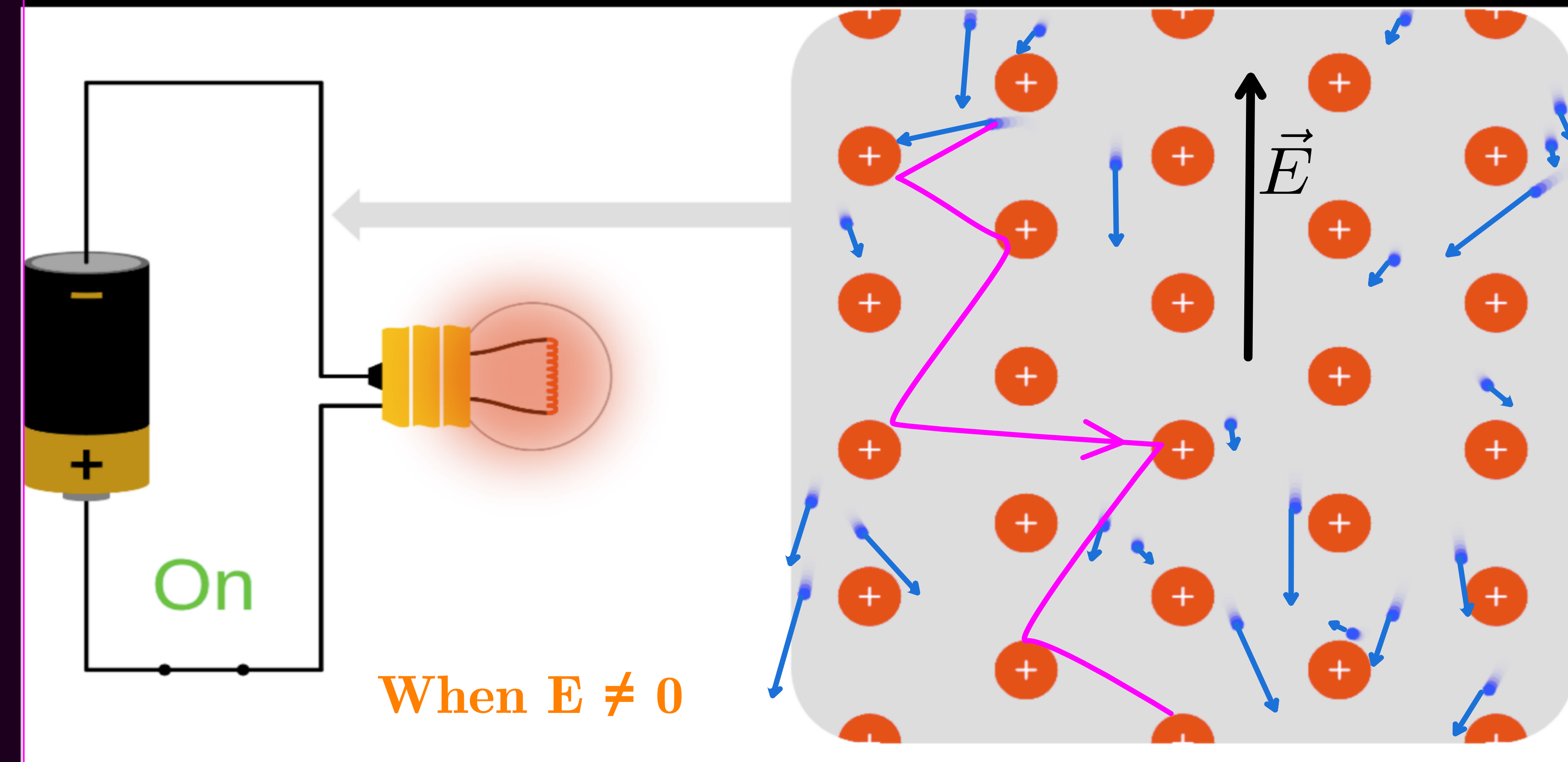




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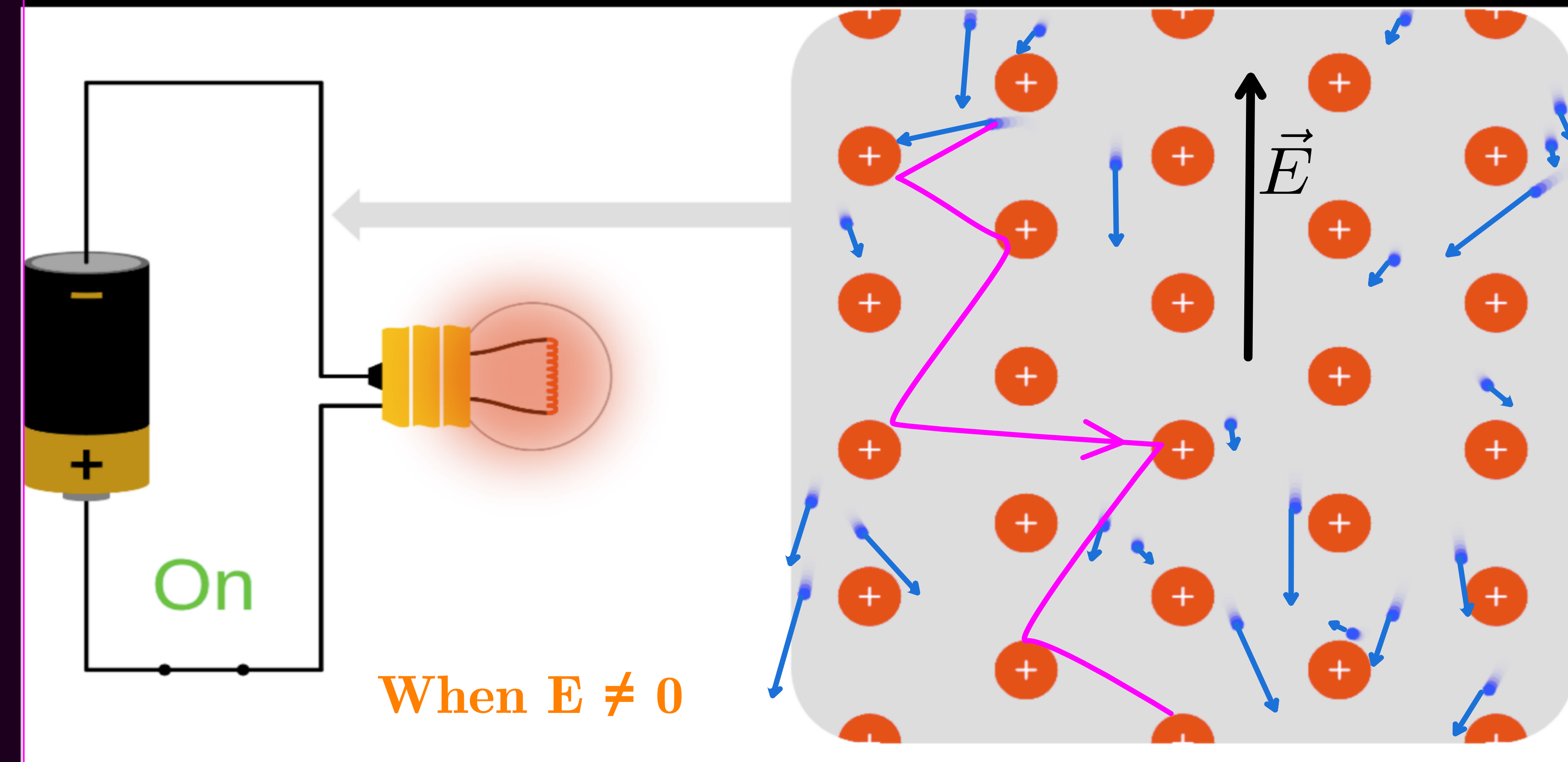




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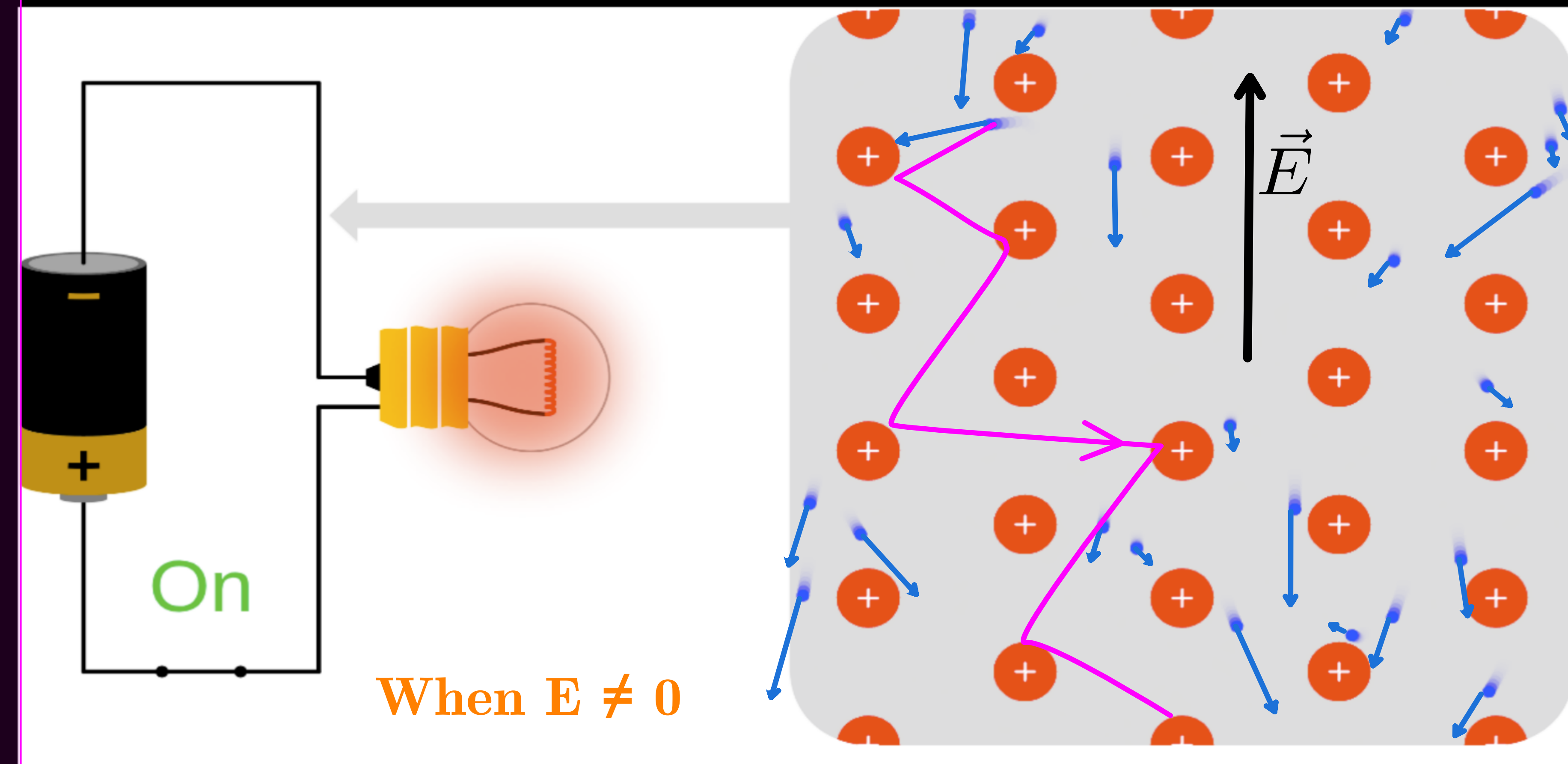




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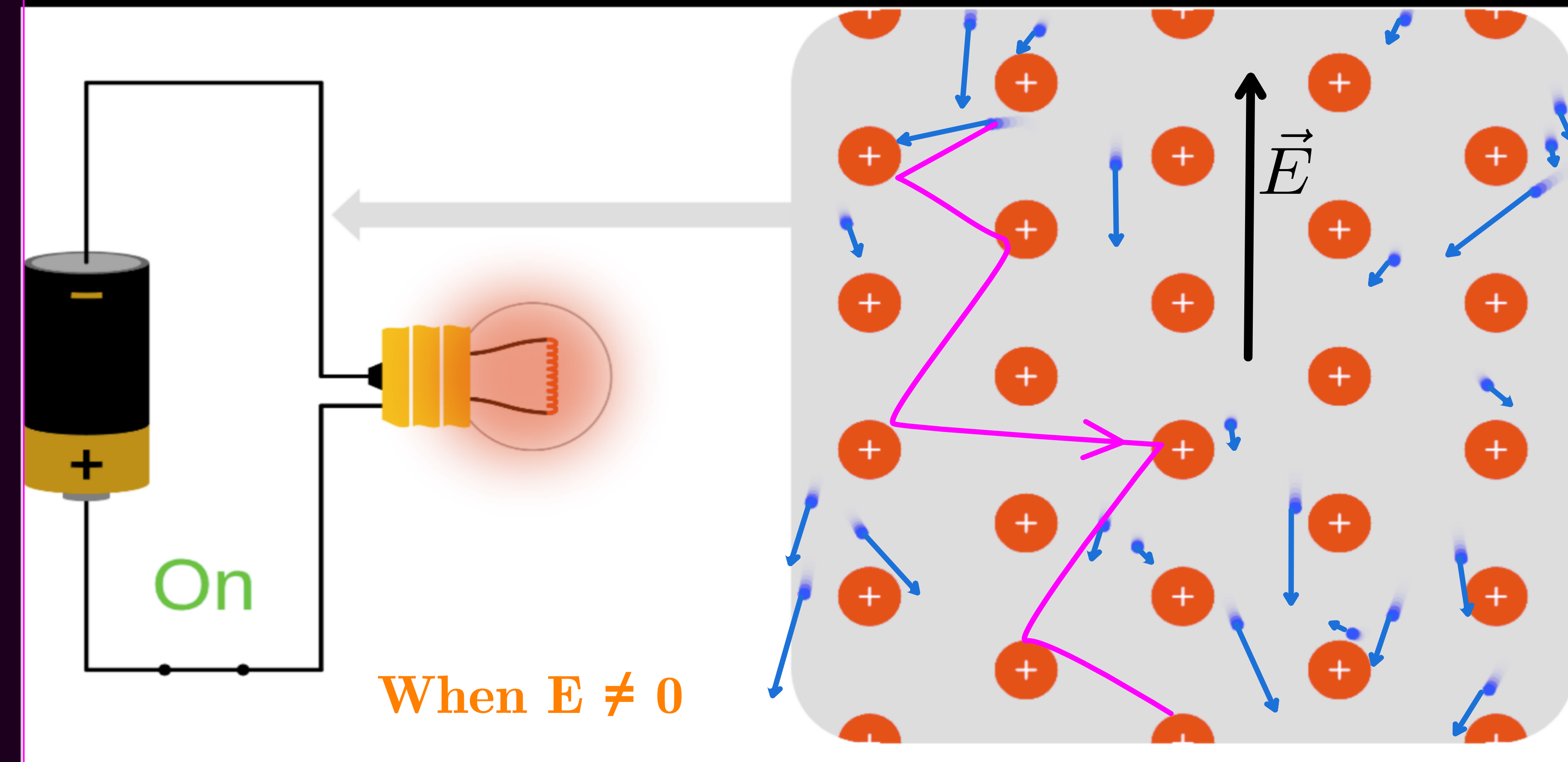
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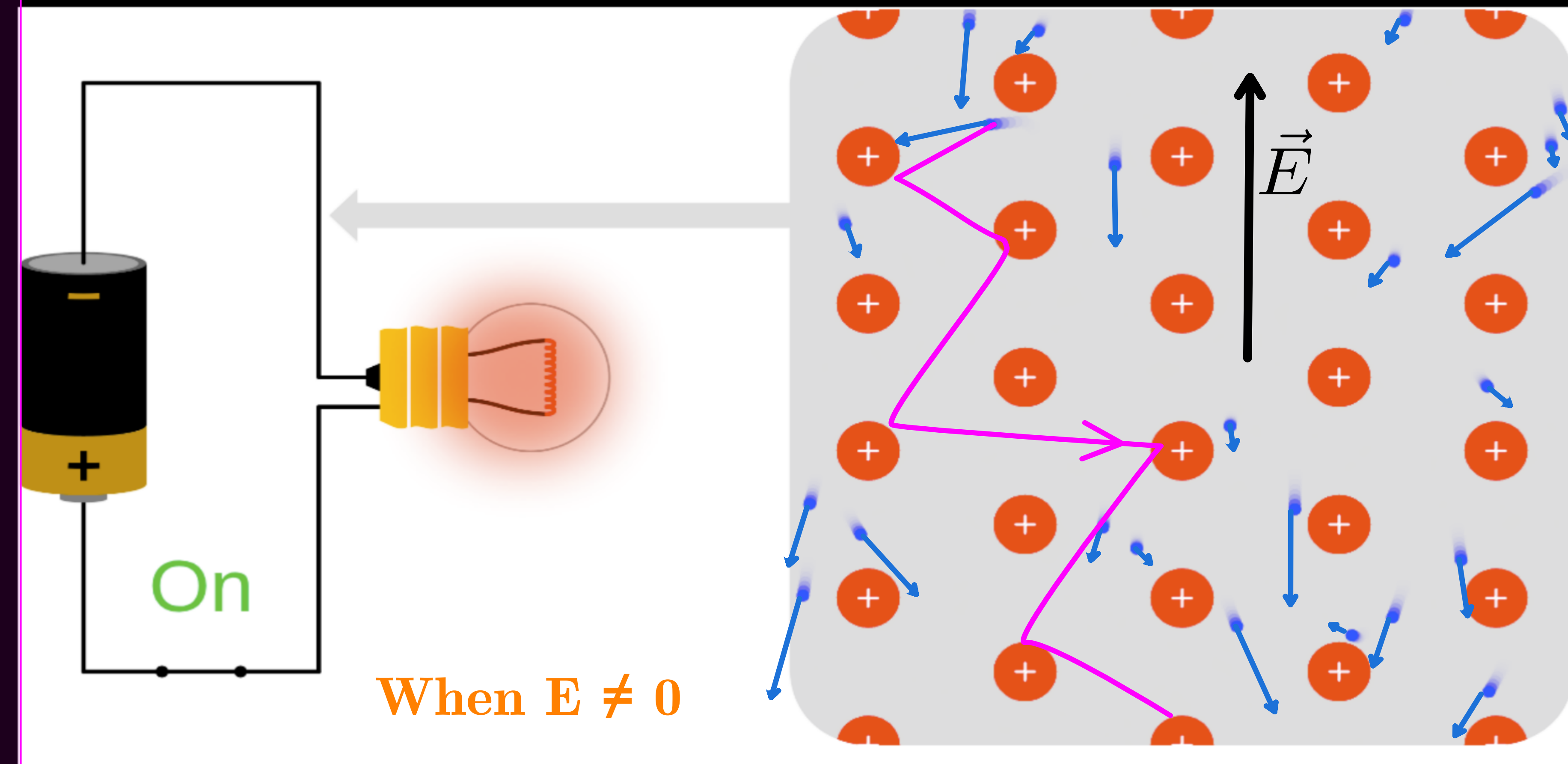
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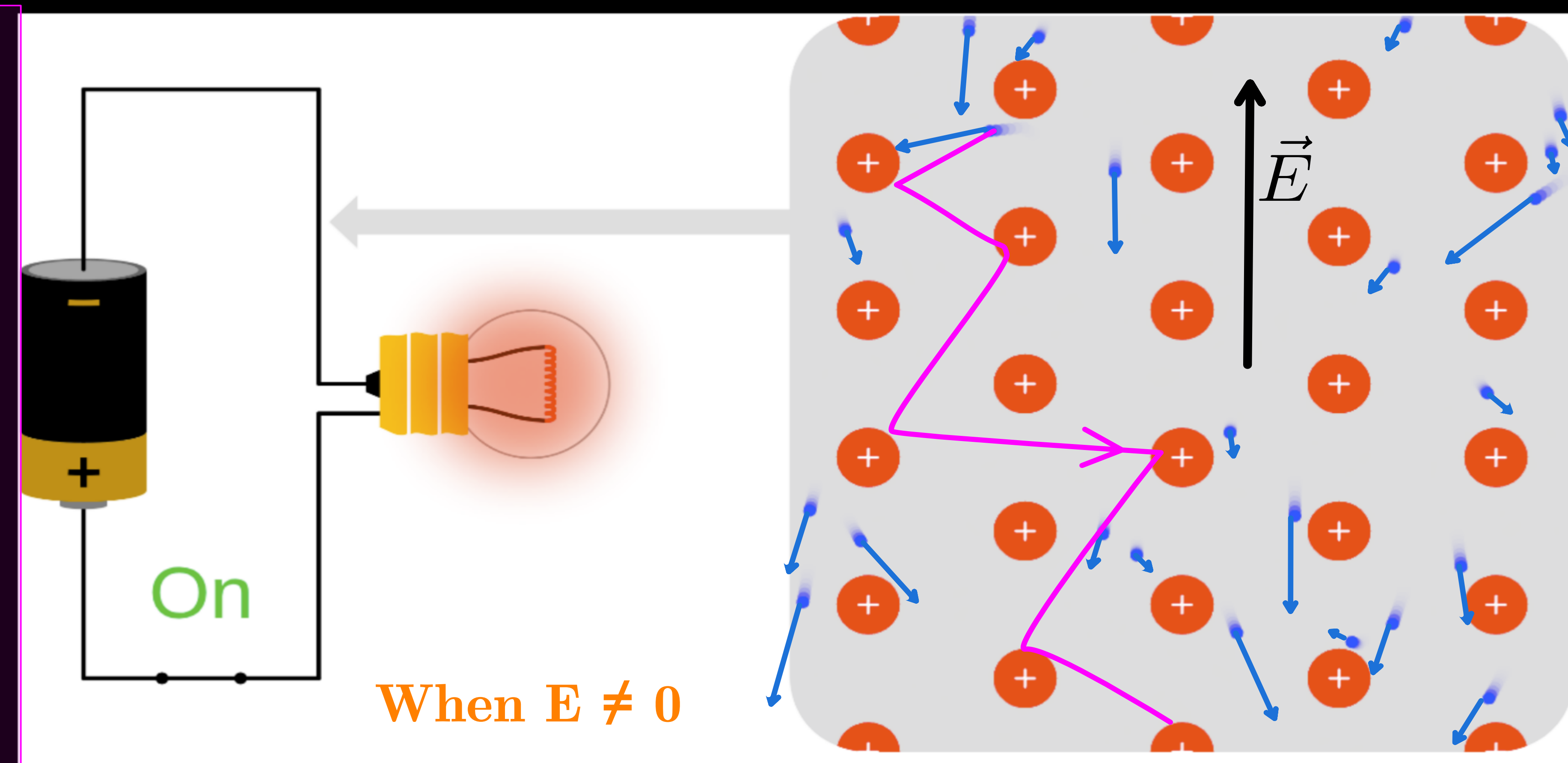


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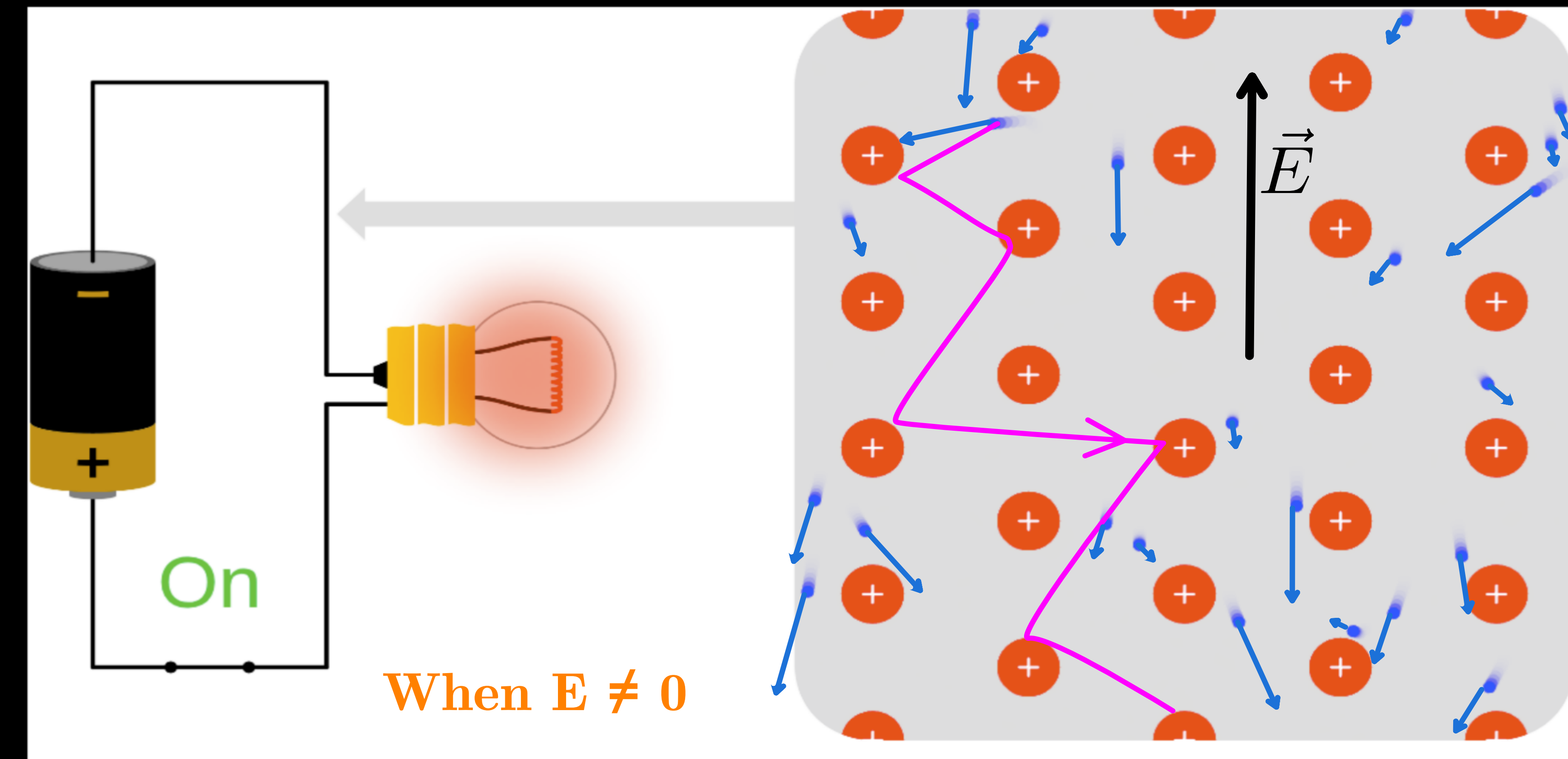
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- \* **Drift velocity** : The average velocity with which the free electrons get drifted in the opposite direction of the applied electric field.



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- \* Due to continuous collisions of electrons and the positive ions, the speed of the electrons doesn't continuously increase. The result of this is that the electrons move within the material with some average constant velocity (drift velocity) in the direction opposite the applied electric field.

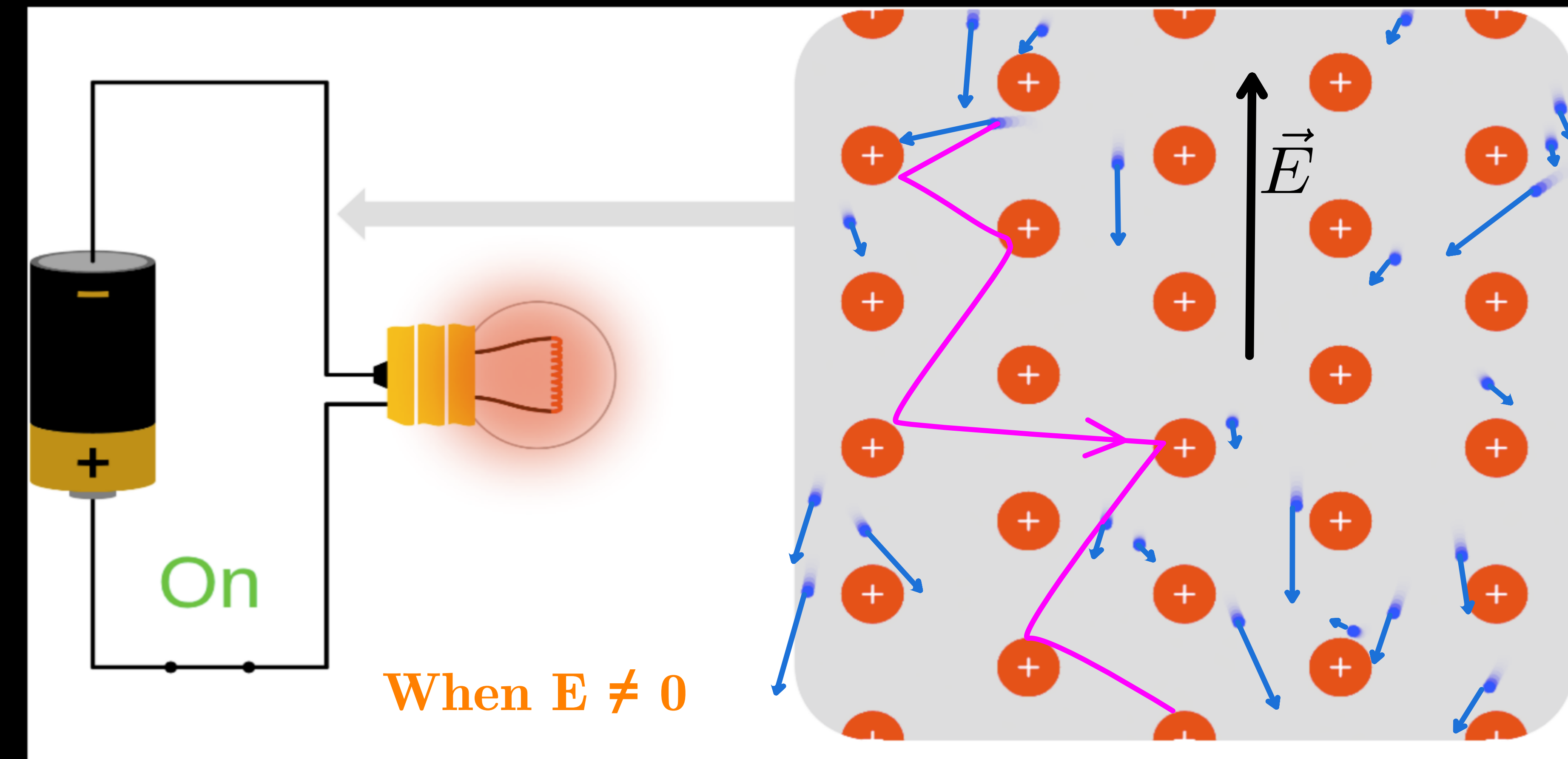




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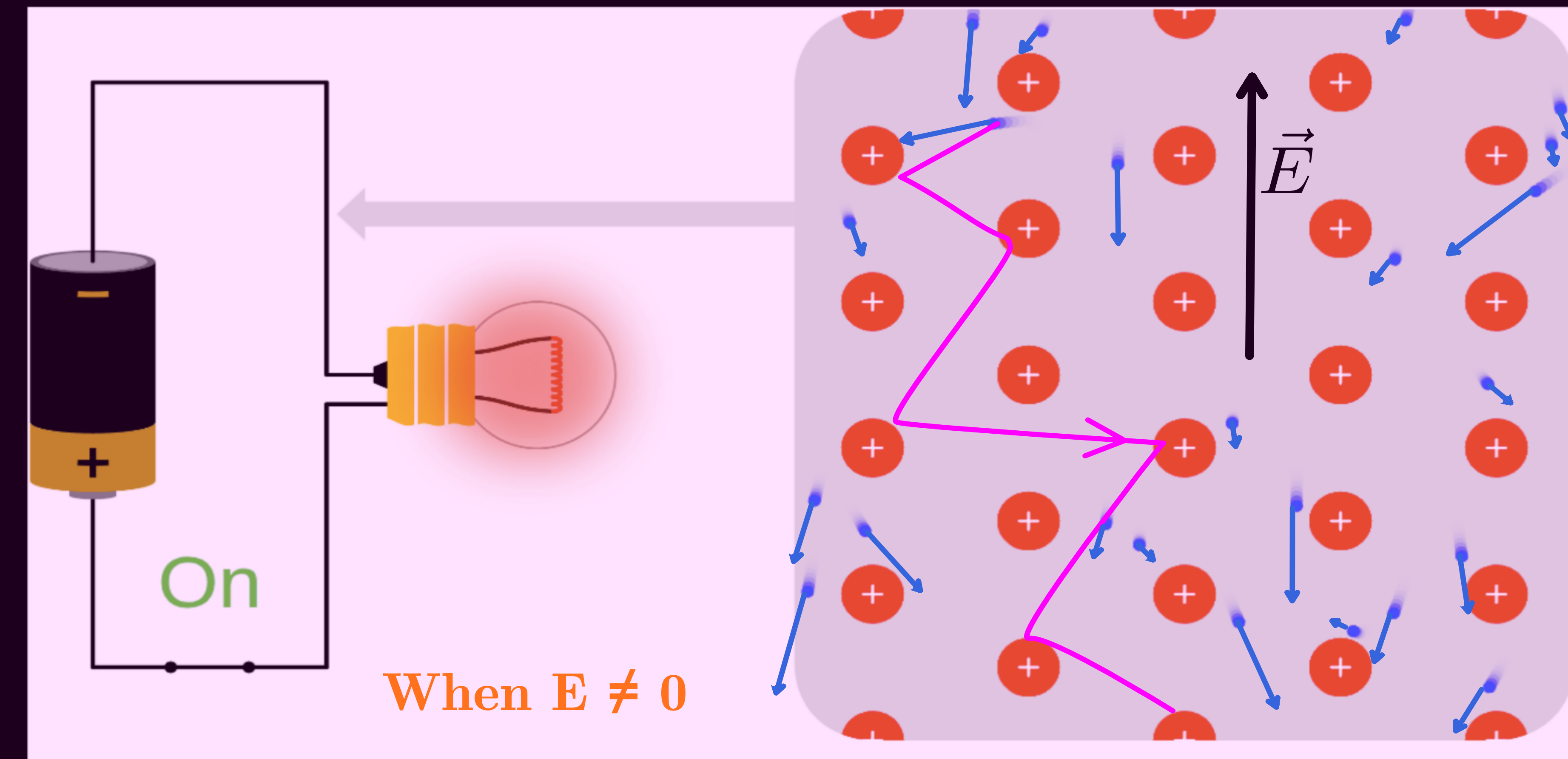


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\* The drift velocity of electrons is very small in the order of  $10^{-4} \text{ ms}^{-1}$ .



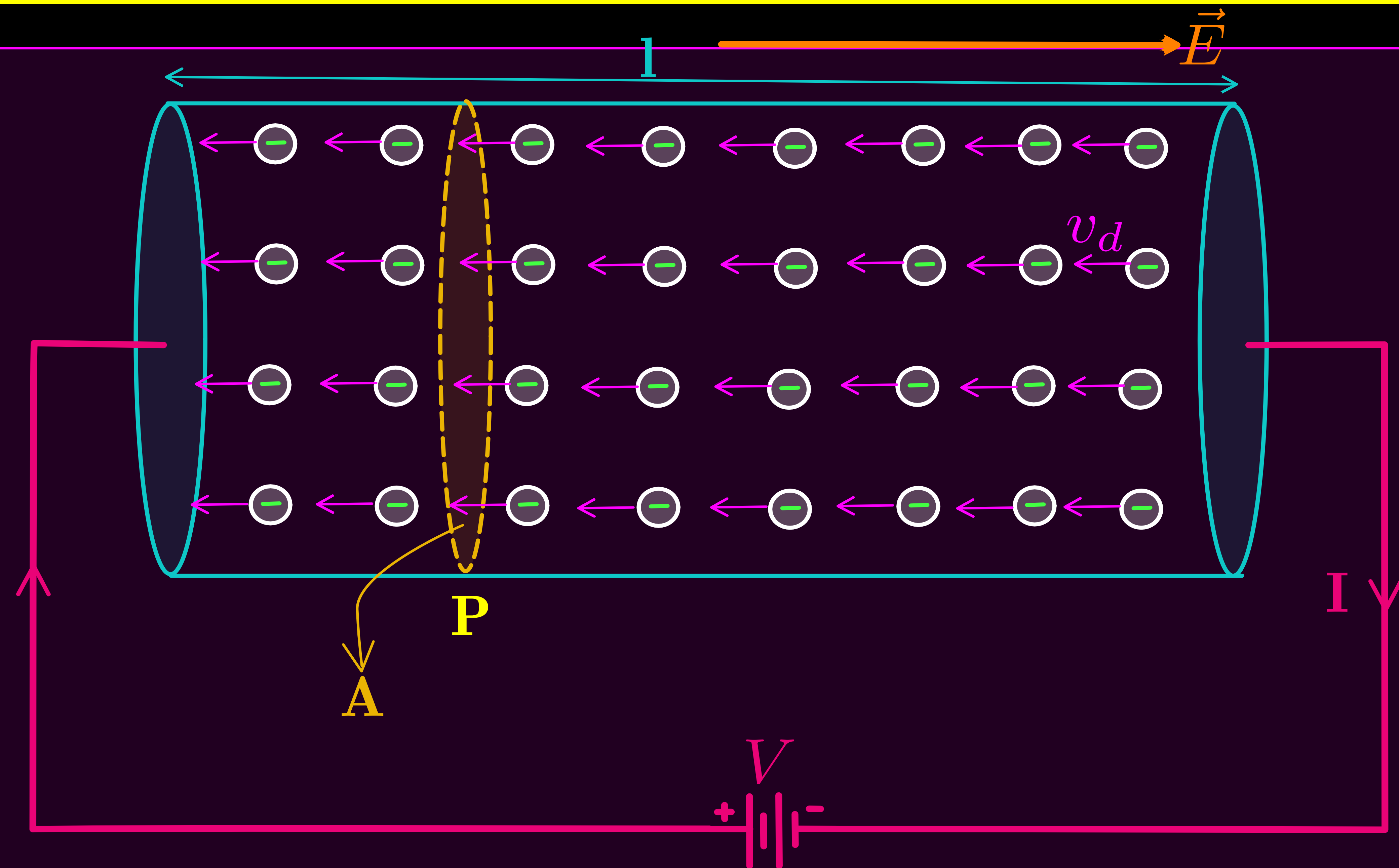


# RELATION BETWEEN CURRENT AND DRIFT VELOCITY



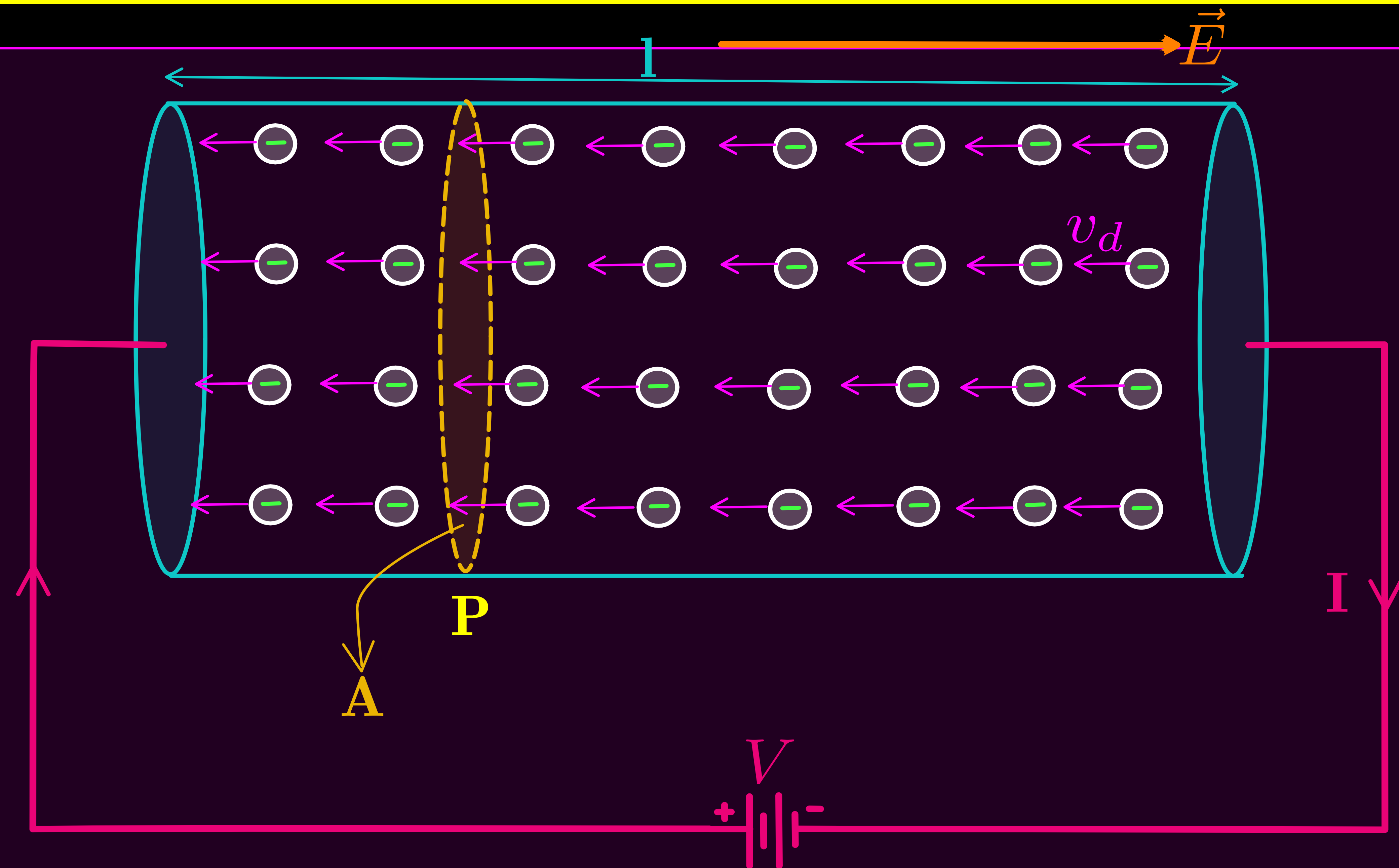


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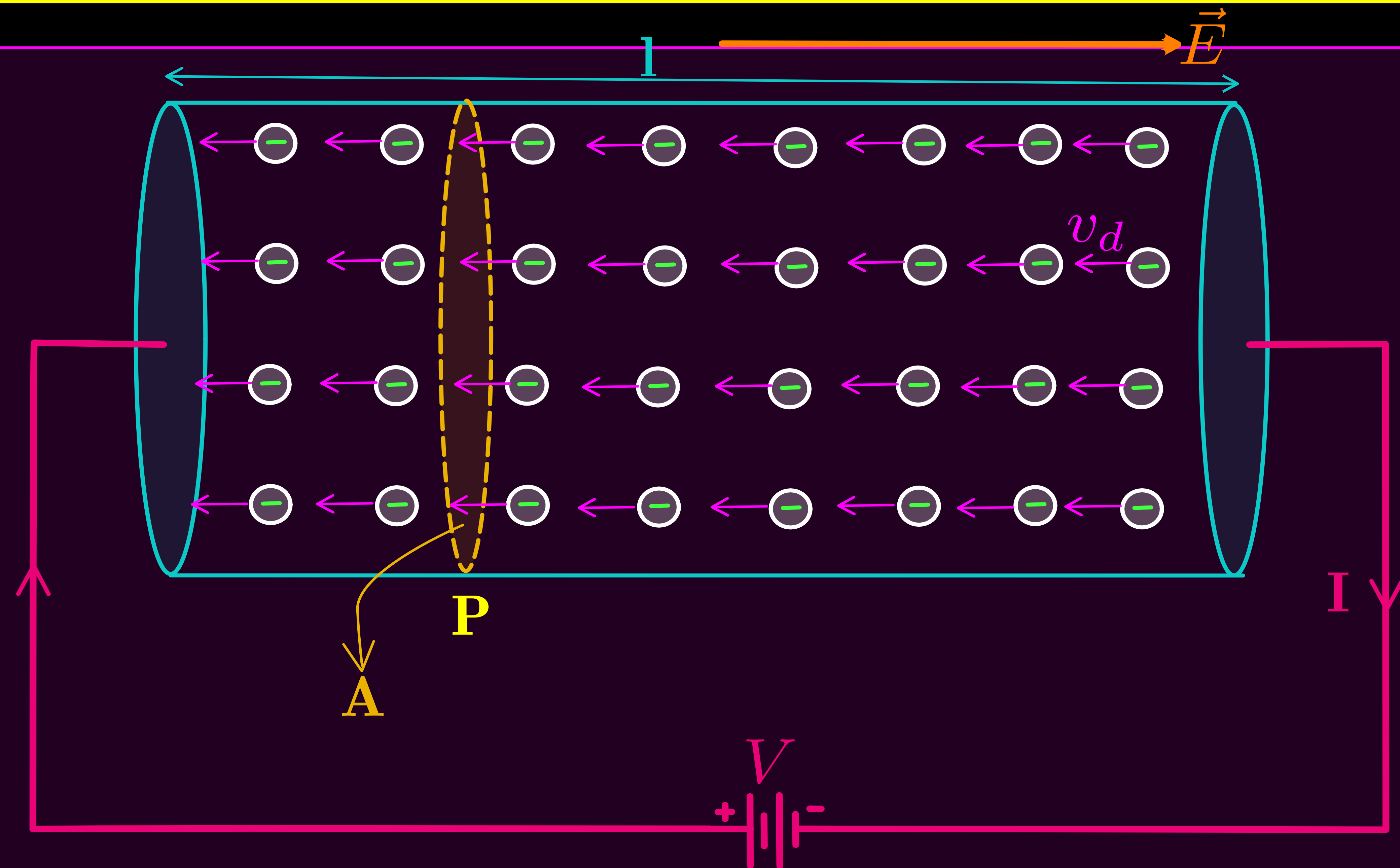
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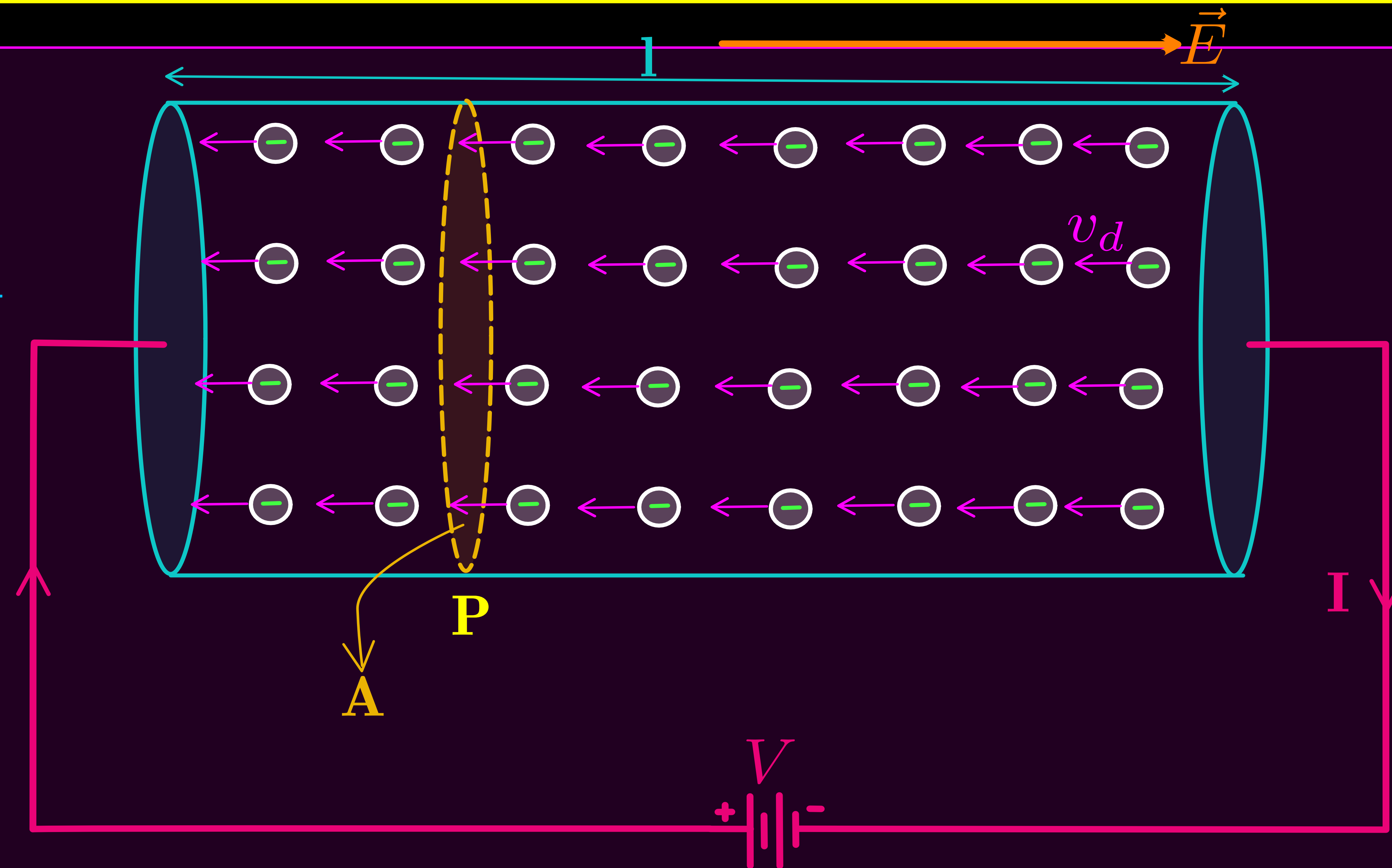
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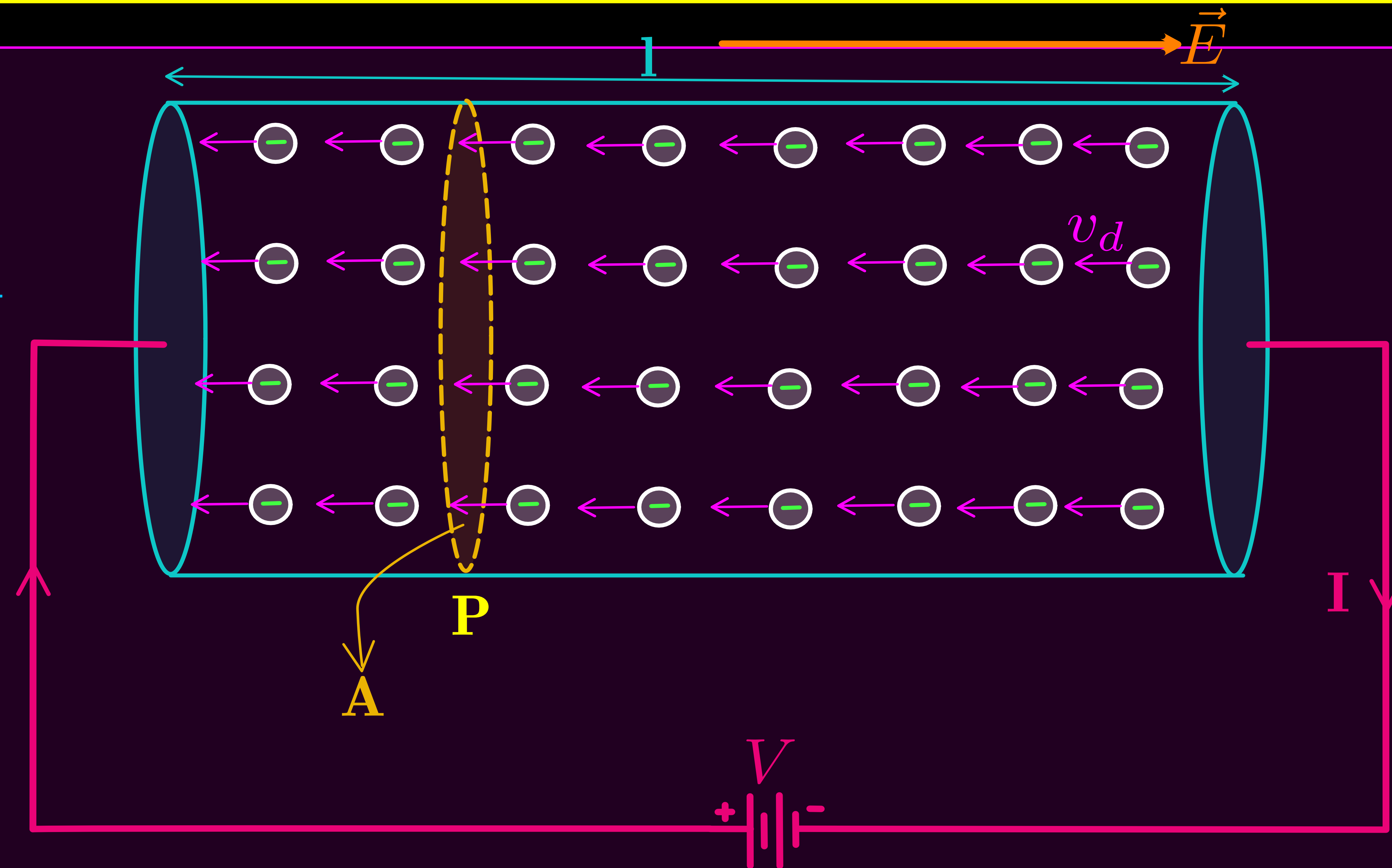
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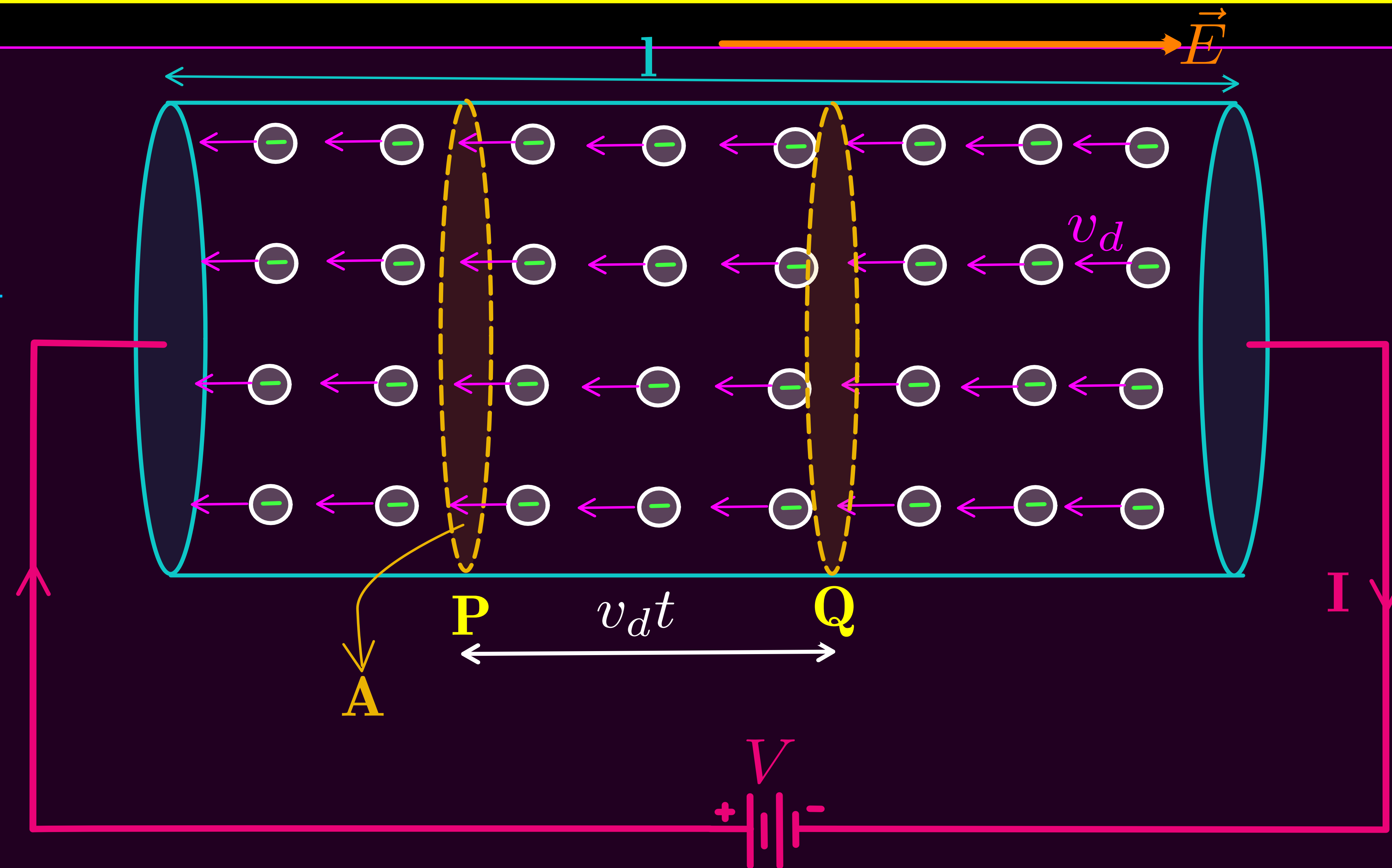
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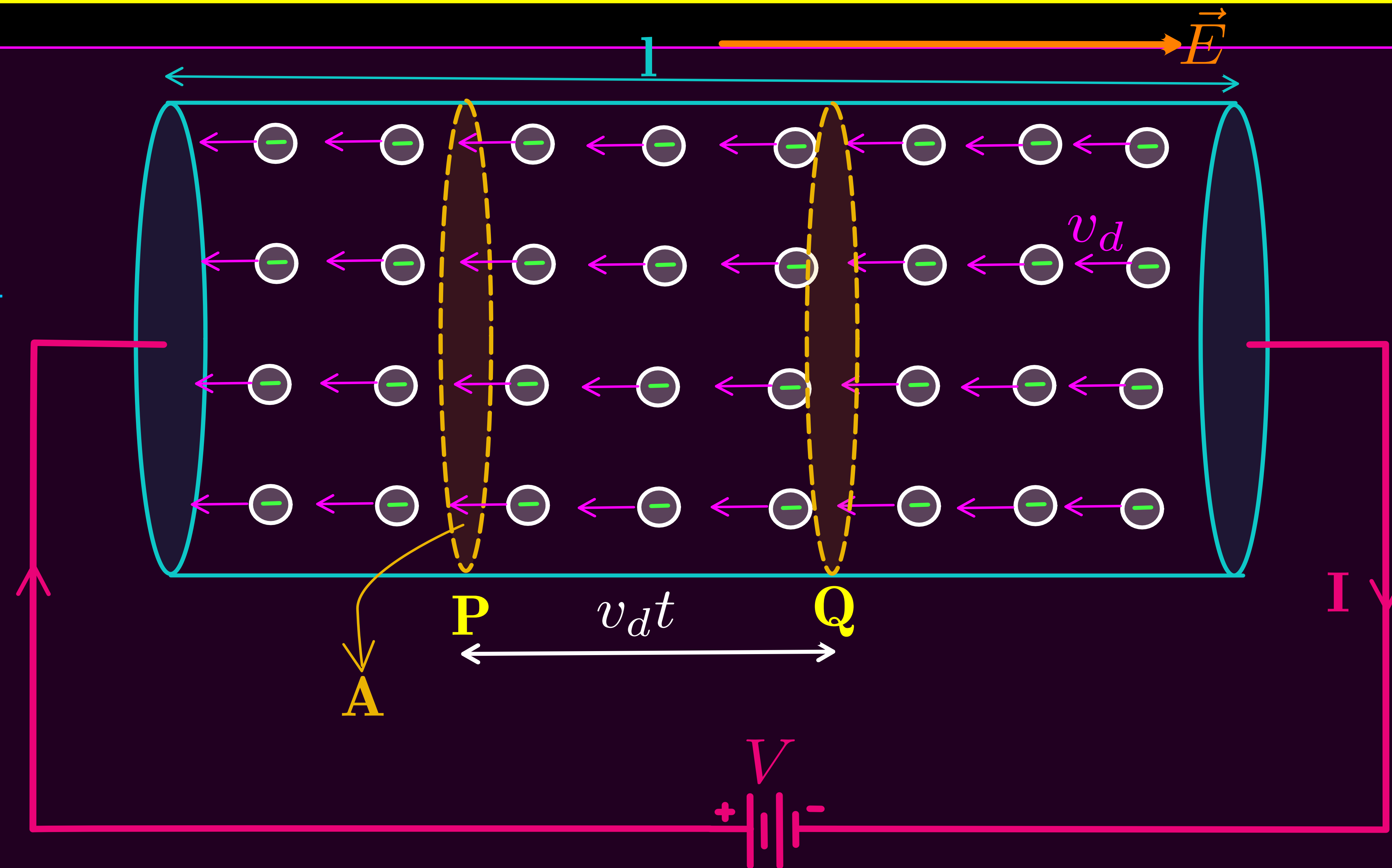
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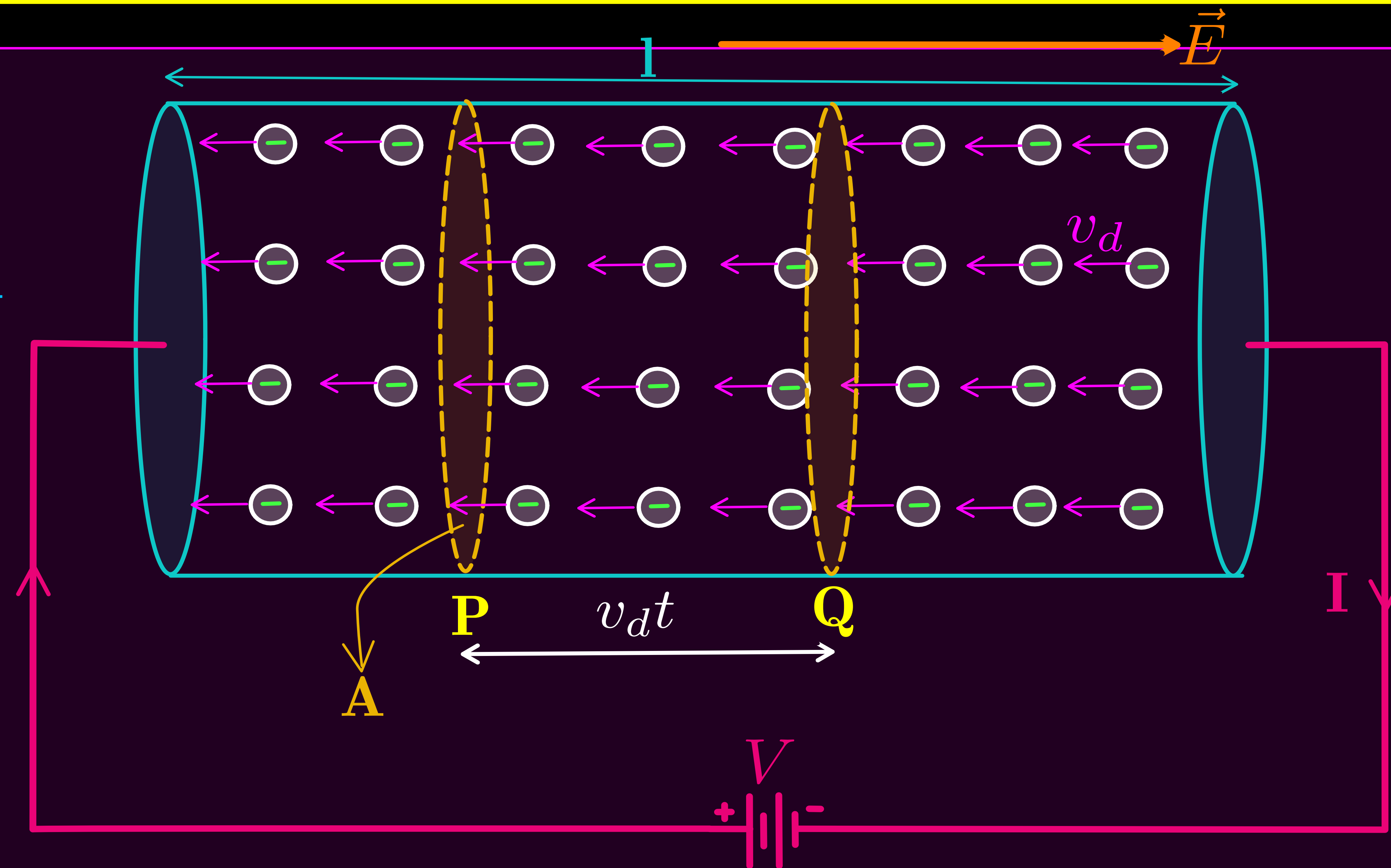
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- \* Therefore, number of electron crossing the plane area P in time  $t = n \times \text{Volume of cylinder PQ}$   

$$= n \times A v_d t$$

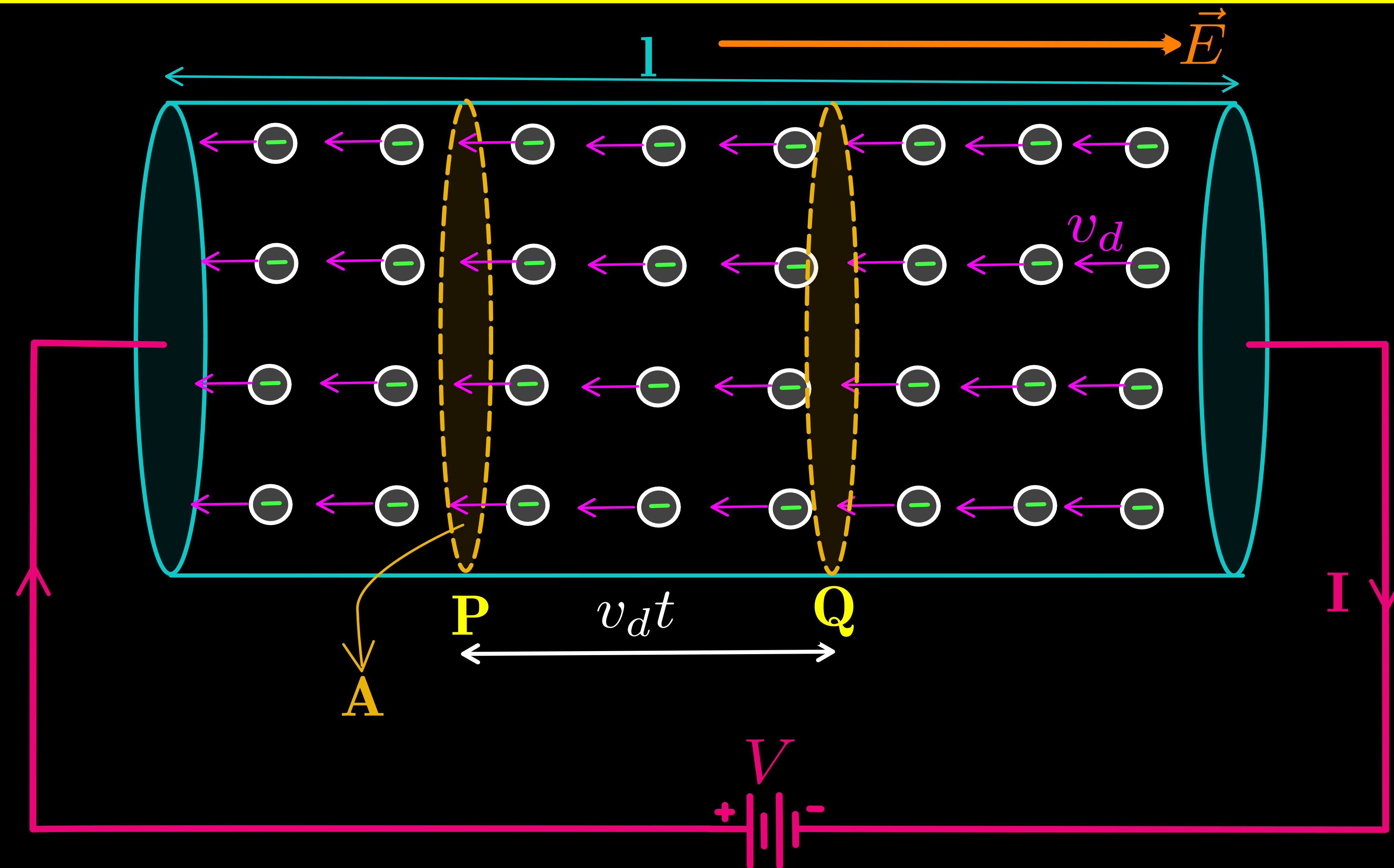


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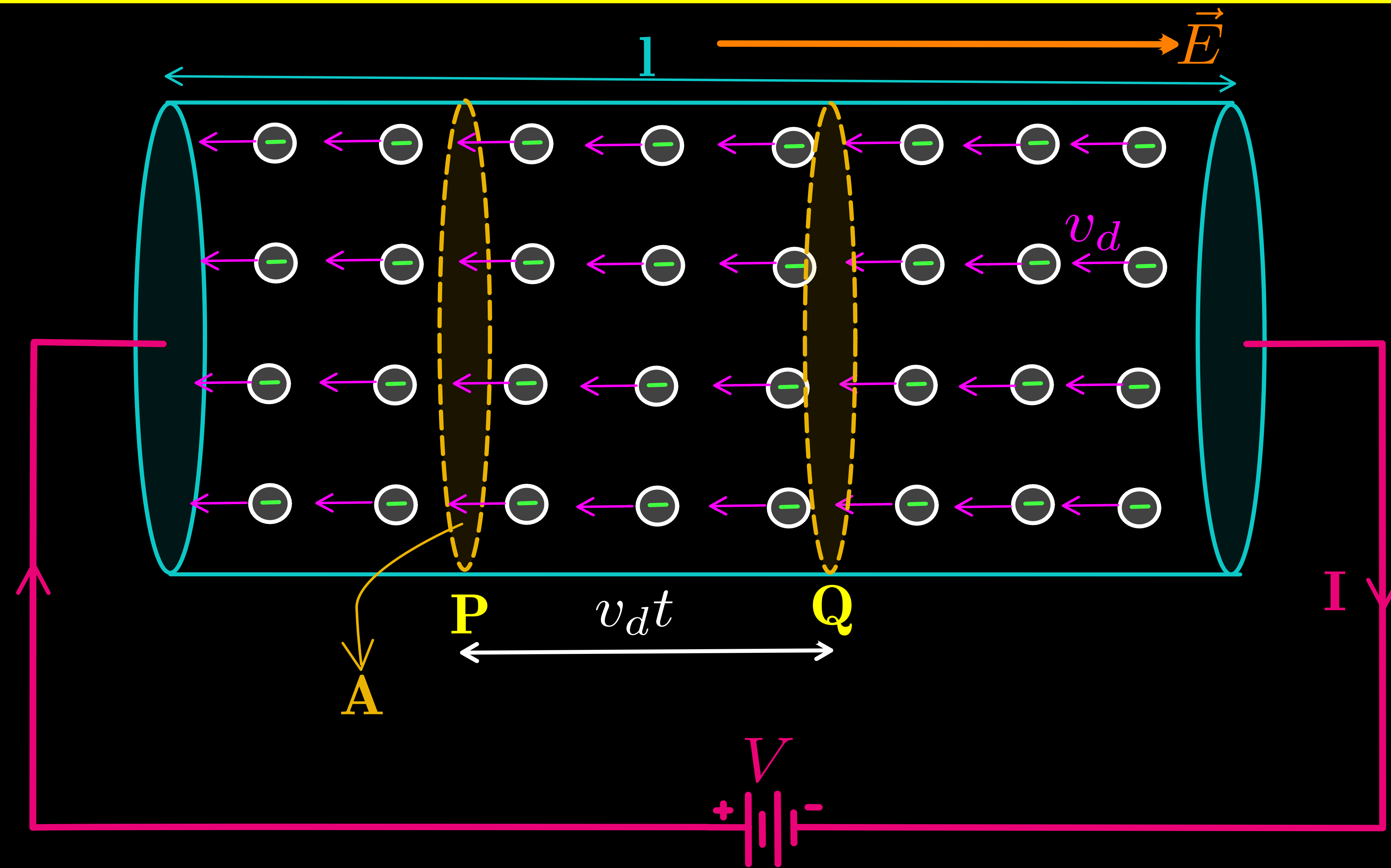
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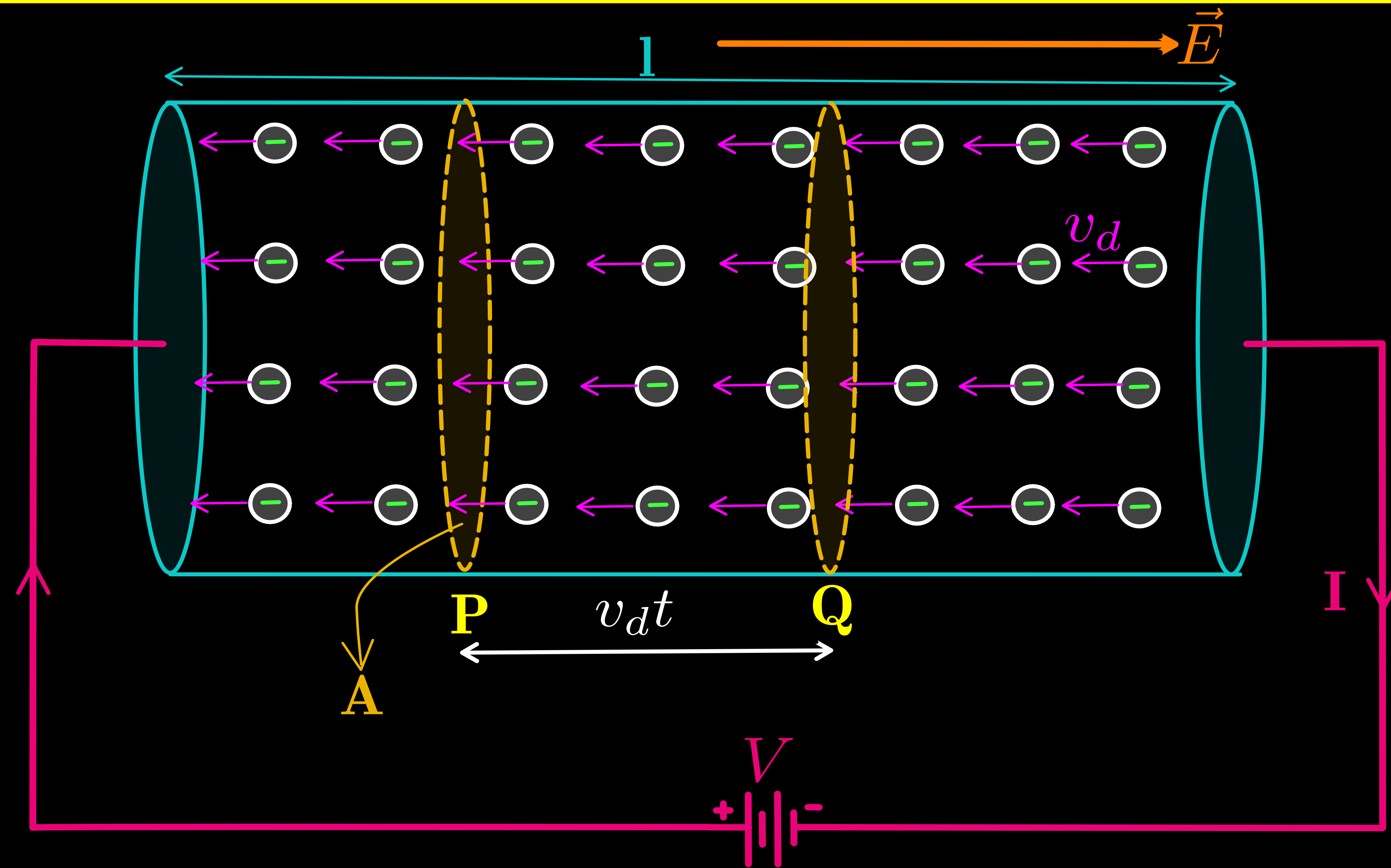
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\* We have already derived that :

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$$\therefore \text{conductivity } \sigma = \frac{ne^2\tau}{m_e}$$

$$\text{and Resistivity } \rho = \frac{1}{\sigma} = \frac{m_e}{ne^2\tau}$$



MOBILITY





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Since,  $v_d = \frac{eE\tau}{m}$

$$\therefore \mu = \frac{v_d}{E} = \frac{e\tau}{m}$$



**Ex 9:** There is a current of 40 A in a wire of  $10^{-6}$  square metre area of cross-section, if the number of free electrons per cubic metre is  $10^{29}$ , then the drift velocity will be

(a)  $1.25 \times 10^3 \text{ m/s}$

(b)  $2.5 \times 10^{-3} \text{ m/s}$

(c)  $25 \times 10^{-3} \text{ m/s}$

(d)  $25 \times 10^3 \text{ m/s}$



**Example 3.1** (a) Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area  $1.0 \times 10^{-7} \text{ m}^2$  carrying a current of 1.5 A. Assume that each copper atom contributes roughly one conduction electron. The density of copper is  $9.0 \times 10^3 \text{ kg/m}^3$ , and its atomic mass is 63.5 u. (b) Compare the drift speed obtained above with, (i) thermal speeds of copper atoms at ordinary temperatures, (ii) speed of propagation of electric field along the conductor which causes the drift motion.



**3. 13** The number density of free electrons in a copper conductor estimated in Example 3.1 is  $8.5 \times 10^{28} \text{ m}^{-3}$ . How long does an electron take to drift from one end of a wire 3.0 m long to its other end? The area of cross-section of the wire is  $2.0 \times 10^{-6} \text{ m}^2$  and it is carrying a current of 3.0 A.



### **Example 3.2**

- (a) In Example 3.1, the electron drift speed is estimated to be only a few  $\text{mm s}^{-1}$  for currents in the range of a few amperes? How then is current established almost the instant a circuit is closed?
- (b) The electron drift arises due to the force experienced by electrons in the electric field inside the conductor. But force should cause acceleration. Why then do the electrons acquire a steady average drift speed?
- (c) If the electron drift speed is so small, and the electron's charge is small, how can we still obtain large amounts of current in a conductor?
- (d) When electrons drift in a metal from lower to higher potential, does it mean that all the 'free' electrons of the metal are moving in the same direction?
- (e) Are the paths of electrons straight lines between successive collisions (with the positive ions of the metal) in the (i) absence of electric field, (ii) presence of electric field?



## **Solution**

- (a) Electric field is established throughout the circuit, almost instantly (with the speed of light) causing at every point a *local electron drift*. Establishment of a current does not have to wait for electrons from one end of the conductor travelling to the other end. However, it does take a little while for the current to reach its steady value.
- (b) Each 'free' electron does accelerate, increasing its drift speed until it collides with a positive ion of the metal. It loses its drift speed after collision but starts to accelerate and increases its drift speed again only to suffer a collision again and so on. On the average, therefore, electrons acquire only a drift speed.
- (c) Simple, because the electron number density is enormous,  $\sim 10^{29} \text{ m}^{-3}$ .
- (d) By no means. The drift velocity is superposed over the large random velocities of electrons.
- (e) In the absence of electric field, the paths are straight lines; in the presence of electric field, the paths are, in general, curved.